

Online Portfolio Optimization with Dynamic-Reference Cumulative Prospect Policy Gradient

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Abstract

Cumulative prospect theory (CPT) models portfolio choice relative to a reference point, yet most first-order CPT policy methods keep that reference fixed even when realized wealth should update it. We study online portfolio optimization with an endogenous reference state, where policy decisions reshape wealth, wealth updates the reference, and the resulting CPT objective moves over time. We introduce Dynamic-Reference Cumulative Prospect Policy Gradient (DRCPT-PG), which combines an exactly unbiased estimator of the regularized frozen-target CPT gradient with finite expected trajectory cost and variance, and a normalized projected update for tracking the moving first-order target. The estimator supports studentized asymptotic inference and attains the fixed-dimensional minimax trajectory-budget rate, while the online analysis controls the projected residual and dynamic local regret under sublinear objective variation. In controlled experiments, continued updating reduces cumulative squared projected residual from 1.427 to 0.684 relative to a frozen closed-loop control. In a chronological 300-stock A-share replay, DRCPT-PG achieves the highest mean Sharpe ratio and the lowest mean maximum drawdown, turnover, and transaction cost among five policy-gradient methods; held-out investor data further show incremental predictive gains from adaptive and asymmetric reference representations. These results connect adaptive reference formation to a statistically controlled first-order learning framework for path-dependent online portfolio optimization.

1 Introduction

Sequential portfolio choice becomes path dependent when wealth is evaluated relative to an adaptive reference point. Two otherwise identical investors can arrive at the same current wealth through different gain–loss histories and nevertheless face different subjective gain/loss states. For example, a path from normalized wealth 1 to 1.4 and then back to 1.2 can leave the investor below an updated reference, whereas a path from 1 to 0.8 and then up to 1.2 can place the same terminal wealth above it. Cumulative prospect theory (CPT) provides a canonical language for such reference-dependent decisions through asymmetric valuation of gains and losses and nonlinear probability weighting [1, 2]. In a sequential portfolio problem, these ingredients make the objective a nonlinear functional of the trajectory–outcome distribution rather than an additive one-period reward.

Recent first-order work has made this distributional objective tractable in reinforcement learning. Foundational studies established prediction and control under CPT [3]; most closely related, Lepel and Barakat [4] derive a policy-gradient theorem for finite-horizon CPT-RL, construct an order-statistics Monte Carlo estimator, and prove convergence and sample-complexity guarantees for first-order stationarity. Their result provides the natural fixed-reference benchmark for the present work. Yet the fixed-reference formulation leaves out a central feature of sequential investing: realized portfolio paths can themselves change the reference against which subsequent gains and losses are judged.

This distinction is supported by behavioral-finance evidence. Security-trading experiments document reference-point adaptation after gains and losses, including asymmetric adjustment [5]. Adaptive and path-dependent references can alter realization and portfolio decisions [6, 7], and recent multi-period portfolio analysis reaches the same conclusion under dynamic loss-aversion mechanisms [8]. Portfolio-wide gain-loss framing also changes disposition behavior [9], while recent evidence from Chinese retail investors shows that the information environment and investor characteristics systematically affect the disposition effect [10]. These findings suggest that the reference point should be treated as part of the evolving decision state rather than as a fixed hyperparameter.

Once the reference becomes endogenous, the difficulty is not merely that the state space grows. The learner’s policy changes realized wealth, realized wealth updates the reference, and the updated reference defines the next CPT objective. This feedback creates two coupled challenges. First, even for a frozen target, the CPT gradient contains nonlinear probability weighting evaluated at an unknown outcome distribution, so a finite-budget plug-in calculation can be biased. Second, the target itself moves through both exogenous market variation and the learner’s own state trajectory. The resulting problem therefore requires both a statistically reliable gradient oracle for each frozen target and an online procedure that tracks a sequence of moving first-order conditions.

We develop *Dynamic-Reference Cumulative Prospect Policy Gradient* (DRCPT-PG) around this decomposition. At a fixed reference state, we construct an exactly unbiased estimator of the regularized CPT gradient with finite expected trajectory cost and variance, together with asymptotic inference and a matching fixed-dimensional minimax budget rate. Across episodes, we use a normalized projected update and analyze its projected residual and dynamic local regret under objective variation and simulator error. A factor-parametrized Dirichlet actor supplies valid long-only portfolio weights without tying the parameter dimension to the number of stocks, while asymmetric reference updating specifies how realized gains and losses reshape the behavioral state.

The paper addresses three linked questions. First, can the nonlinear CPT policy gradient be estimated without finite-sample bias while retaining usable variance and trajectory-budget guarantees? Second, under what variation and simulation-error conditions can online updating keep the projected first-order residual small as the market and reference state evolve? Third, does the same dynamic-reference mechanism remain visible when moving from controlled experiments to chronological A-share portfolios and held-out investor behavior? Together, these questions connect the statistical, optimization, and behavioral parts of the model: frozen-target estimation supplies a reliable local oracle, online tracking propagates it through the moving objective, and downstream evaluations test whether the reference mechanism remains informative beyond the controlled setting.

The results follow this sequence. Controlled experiments first establish path dependence, finite-sample gradient behavior, and moving-target tracking, and then show how asymmetric reference adaptation changes the strength of the tracking advantage. A chronological 300-stock A-share replay evaluates the method under point-in-time market data, where DRCPT-PG attains the highest mean Sharpe ratio together with the lowest mean maximum drawdown, turnover, and transaction cost among the five policy-gradient methods. Held-out investor decisions show that introducing a reference produces the main predictive gain, with adaptive and asymmetric updating adding further information about realized trading behavior. A final calibrated heterogeneous-agent replay examines how these differences carry into repeated recommendation interactions.

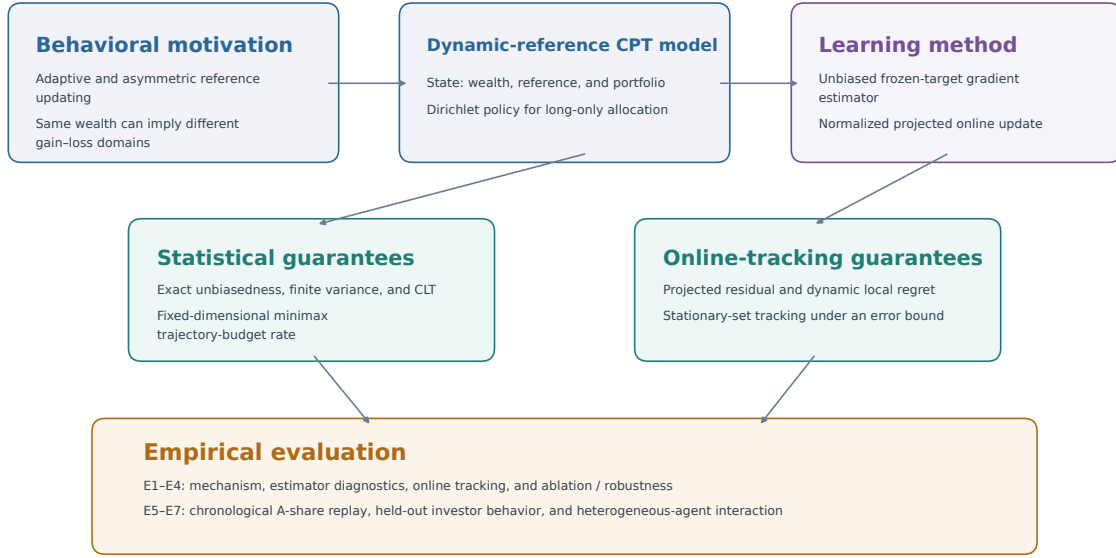


Figure 1: Overall research framework of DRCPT-PG. Adaptive reference formation motivates a dynamic-reference CPT portfolio model. The learning layer combines a debiased frozen-target gradient estimator with normalized projected online updates, leading to statistical and tracking guarantees that are evaluated in controlled, market, and behavioral experiments.

Contributions.

1. **Endogenous-reference CPT policy optimization.** We formulate a finite-horizon portfolio objective in which the psychological reference point evolves with realized wealth. This turns fixed-reference CPT optimization into a path-dependent moving-target problem while retaining the fixed-reference case as a special case.
2. **Debiased gradient estimation with statistical guarantees.** We establish the regularized CPT score-gradient identity and construct a centered leave-one-out estimator with an independent randomized correction. The resulting estimator is exactly unbiased, has finite expected cost and finite variance, obeys a multivariate central limit theorem, and attains the fixed-dimensional minimax rate under the same expected complete-trajectory budget.
3. **Online first-order tracking for a moving nonconvex target.** We analyze a normalized projected update through the projected residual. The resulting guarantee separates objective variation, replication error, and frozen-simulator error; it implies a dynamic-local-regret bound and, under a local error-bound geometry, tracking of the moving stationary set.
4. **Cross-setting evaluation of dynamic reference formation.** Controlled experiments evaluate the estimator and moving-target tracking mechanism directly. A chronological A-share replay and held-out investor decisions then test whether dynamic references remain useful in market and behavioral settings, with a calibrated heterogeneous-agent replay providing an additional interaction-level evaluation.

Organization. Section 2 positions the paper relative to CPT optimization, adaptive-reference finance, nonstationary nonconvex learning, statistical debiasing, and recent portfolio and investor-facing financial systems. Section 3 defines the dynamic-reference portfolio objective and projected tracking criterion, Section 4 presents the debiased estimator and online update, and Section 5 develops the frozen-target statistical guarantees before the moving-target tracking results. Section 6 follows the evidence from controlled mechanism and estimator tests to market, behavioral, and interaction evaluation, and Section 7 synthesizes the resulting implications and connections to prior work. Proofs are collected in the appendix.

2 Related Work and Positioning

CPT optimization and reinforcement learning. Prospect theory and CPT provide the behavioral foundations for reference-dependent valuation and nonlinear probability weighting [1, 2]. Their use in sequential learning began with CPT prediction and control in reinforcement learning [3], while recent work has made the optimization side substantially more explicit. Lepel and Barakat [4] derive a first-order policy-gradient theorem for finite-horizon CPT-RL, together with an order-statistics Monte Carlo estimator and convergence and sample-complexity guarantees for approximate stationarity. Static and portfolio-specific CPT optimization has also advanced through convex-analytic and splitting-based formulations [11–13]. These studies establish that CPT objectives can be optimized despite their nonconvex and distribution-dependent structure. Our setting departs from them at the level of the target itself: realized wealth changes the reference state, so the policy faces a path-dependent sequence of CPT objectives rather than repeated optimization of one fixed reference-dependent criterion.

Adaptive references, path dependence, and investor behavior. The empirical motivation for an evolving reference predates the recent optimization literature. Security-trading experiments document reference-point adaptation after gains and losses, including asymmetric adjustment [5]. Subsequent models show that adaptive references can alter realization behavior [6], while path-dependent references make current decisions depend on the route by which wealth reaches its present level [7]. More recent portfolio analysis embeds dynamic reference mechanisms in multi-period allocation [8]; complementary evidence shows that disposition behavior depends on portfolio-wide gain–loss frames [9] and, for Chinese retail investors, on the information environment and investor characteristics [10]. Together, this literature explains why the reference should evolve with the realized portfolio path. What it does not yet provide is a first-order CPT policy-gradient framework whose gradient estimation and online tracking theory explicitly account for that endogenous reference evolution.

Nonstationary and nonconvex online optimization. When objectives are nonconvex and time varying, global comparator regret is often too strong to describe first-order learning. Dynamic local regret instead evaluates stationarity after smoothing recent first-order information [14]. Recent work has sharpened this viewpoint through gradient-mapping criteria for online nonconvex composite optimization [15] and projected-gradient analyses for constrained nonconvex and stochastic smooth problems [16]. In nonstationary reinforcement learning, variation budgets likewise quantify how much environmental drift can be tolerated while retaining learnability [17]. Our problem shares the need for a local first-order criterion but differs in the origin of nonstationarity: part of the objective drift is generated by the learner’s own wealth path and reference recursion. We therefore work with

the normalized projected residual of the contemporaneous CPT objective and derive dynamic local regret from its temporally smoothed trajectory.

Debiasing and statistical guarantees for the gradient oracle. For a fixed reference, the CPT policy gradient remains statistically unusual because the policy score is multiplied by a nonlinear functional of an unknown survival distribution. The order-statistics estimator of Lepel and Barakat [4] is the closest first-order baseline and comes with consistency and optimization guarantees. We instead ask for an exact finite-sample property of the frozen-target gradient oracle. Building on randomized unbiased telescoping ideas [18], we combine a centered leave-one-out base with a randomized full/half correction and obtain exact unbiasedness together with finite expected trajectory cost and variance. The same construction supports studentized asymptotic inference and a fixed-dimensional minimax trajectory-budget characterization, which makes the oracle directly usable inside the online tracking analysis.

Portfolio evaluation and investor-facing systems. The downstream experiments connect the theoretical mechanism to two empirical literatures. Dirichlet factor policies provide a natural precedent for simplex-valued reinforced portfolios whose concentration parameters depend on firm characteristics [19], while recent financial-RL work emphasizes time-varying market environments [20]. Evidence on Chinese A-shares also documents market-specific factor structure [21], motivating a chronological point-in-time replay rather than a static random split. A separate line of work studies persona-specific economic risk preferences in language models [22] and personalized generative-AI financial advisers [23]. Our behavioral evaluation complements these studies by asking whether dynamic reference states add information about held-out investor decisions and how reference-aware portfolio recommendations interact with heterogeneous investor profiles.

3 Problem Setup

The dynamic-reference formulation couples three objects that would be separate in a fixed-reference portfolio model: the market and wealth process, the reference state induced by the realized path, and the CPT objective optimized by the policy. We first define the episodic portfolio dynamics and reference recursion, then specify the Dirichlet policy and regularized CPT objective, and finally introduce the variation and first-order tracking quantities used by the online analysis.

3.1 Nonstationary episodic portfolio process

Episodes are indexed by $k = 1, \dots, K$, begin at dates t_k , and have a fixed horizon $h \geq 1$. For consecutive episodes, $t_{k+1} = t_k + h$. At calendar time $u \in \{t_k, \dots, t_k + h - 1\}$,

$$s_u = (m_u, X_u, r_u, \omega_{u-1}), \quad \omega_u \in \Delta_N := \left\{ \omega \in \mathbb{R}_+^{N+1} : \sum_{i=0}^N \omega_i = 1 \right\}, \quad (1)$$

where asset 0 is cash, $X_u > 0$ is normalized wealth, and $r_u \geq 0$ is the reference wealth. For return vector R_{u+1} with $R_{0,u+1} = 0$ and proportional transaction-cost coefficient $c_{tc} \geq 0$,

$$X_{u+1} = X_u \left(1 + \omega_u^\top R_{u+1} - c_{tc} \|\omega_u - \omega_{u-1}\|_1 \right), \quad X_1 = 1. \quad (2)$$

The conditional market transition law may vary with calendar time. The likelihood-structure restriction used for policy-gradient differentiation is stated separately in Assumption 3.2.

3.2 Dynamic reference and terminal relative outcome

For fixed $\eta_+, \eta_- \in [0, 1)$,

$$r_{u+1} = \Phi(r_u, X_{u+1}) = \begin{cases} (1 - \eta_+)r_u + \eta_+X_{u+1}, & X_{u+1} \geq r_u, \\ (1 - \eta_-)r_u + \eta_-X_{u+1}, & X_{u+1} < r_u. \end{cases} \quad (3)$$

Let $\mathcal{F}_k$ be the information available at the start of episode k , and set

$$r_k := r_{t_k}, \quad z_k := (m_{t_k}, X_{t_k}, \omega_{t_k-1}). \quad (4)$$

All Monte Carlo learning trajectories for episode k are generated conditionally on the frozen context $(\mathcal{F}_k, z_k, r_k)$. Each trajectory τ carries its own wealth and reference recursions and produces

$$Y_k(\tau) = X_{t_k+h}(\tau) - r_{t_k+h}(\tau). \quad (5)$$

3.3 Dirichlet portfolio policy

For asset $i = 0, \dots, N$, let $q_{i,u} \in \mathbb{R}^d$ be a feature vector. For $\theta \in \Theta \subset \mathbb{R}^d$ define

$$a_{i,u}(\theta) = \exp(\theta^\top q_{i,u}), \quad \tilde{\omega}_u \sim \pi_\theta(\cdot | s_u) = \text{Dirichlet}(a_{0,u}(\theta), \dots, a_{N,u}(\theta)). \quad (6)$$

A history-measurable feasibility map $M_u(\cdot | s_u) : \Delta_N \rightarrow \Delta_N$ and a fixed trade fraction $\chi \in (0, 1]$ generate the executed portfolio

$$\omega_u = (1 - \chi)\omega_{u-1} + \chi M_u(\tilde{\omega}_u | s_u). \quad (7)$$

The realized path and all learning trajectories use the same stochastic actor (6) and the same map (7). The augmented trajectory records $\tilde{\omega}_u$, so its score is

$$G_{k,\theta}(\tau) = \sum_{u=t_k}^{t_k+h-1} \sum_{i=0}^N a_{i,u} q_{i,u} \left\{ \log \tilde{\omega}_{i,u} - \psi(a_{i,u}) + \psi \left(\sum_{j=0}^N a_{j,u} \right) \right\}, \quad (8)$$

where ψ is the digamma function.

3.4 Regularized CPT objective

For $x \geq 0$ let

$$u_{\alpha, \varepsilon_v}(x) = (x^2 + \varepsilon_v^2)^{\alpha/2} - \varepsilon_v^\alpha, \quad v_+(y) = u_{\alpha_+, \varepsilon_v}(y^+), \quad v_-(y) = \lambda u_{\alpha_-, \varepsilon_v}(y^-), \quad (9)$$

with $0 < \alpha_+, \alpha_- \leq 1$, $\lambda > 1$, and $\varepsilon_v > 0$. Let $w_\pm$ be strictly increasing probability-weighting functions and, for fixed $\varepsilon \in (0, 1/2)$, define the endpoint-normalized weights

$$w_\pm^\varepsilon(p) = \frac{w_\pm(\varepsilon + (1 - 2\varepsilon)p) - w_\pm(\varepsilon)}{w_\pm(1 - \varepsilon) - w_\pm(\varepsilon)}, \quad 0 \leq p \leq 1. \quad (10)$$

Thus $w_\pm^\varepsilon(0) = 0$ and $w_\pm^\varepsilon(1) = 1$.

For a frozen episode environment/model P and context (z, r) , let P_θ denote the induced augmented trajectory law under π_θ . Write

$$S_{P,z,r,\theta}^\pm(y) = P_\theta\{v_\pm(Y) > y\}.$$

Under Assumption 3.1, $v_{\pm}(Y) \in [0, B_v]$ on the analyzed domain. Define

$$J(P, z, r; \theta) = \int_0^{B_v} w_+^\varepsilon \left(S_{P,z,r,\theta}^+(y) \right) dy - \int_0^{B_v} w_-^\varepsilon \left(S_{P,z,r,\theta}^-(y) \right) dy. \quad (11)$$

Let P_k denote the true conditional episode- k model and $\tilde{P}_k$ its frozen simulator model. For fixed-target analysis, the episode context and model are conditioned upon before fresh trajectories are drawn; Assumption 3.2 specifies how θ enters the resulting trajectory law. Then

$$J_k(\theta, r_k) := J(P_k, z_k, r_k; \theta), \quad \tilde{J}_k(\theta, r_k) := J(\tilde{P}_k, z_k, r_k; \theta), \quad (12)$$

with gradients

$$g_k(\theta) := \nabla_\theta J_k(\theta, r_k), \quad \tilde{g}_k(\theta) := \nabla_\theta \tilde{J}_k(\theta, r_k). \quad (13)$$

3.5 Online approximation and variation budgets

At the realized iterate, define the frozen-simulator gradient discrepancy

$$\beta_k := \|\tilde{g}_k(\theta_k) - g_k(\theta_k)\|_2, \quad (14)$$

and the cumulative variation of the true episode objectives

$$\mathcal{V}_K := \sum_{k=1}^{K-1} \sup_{\theta \in \Theta} |J_{k+1}(\theta, r_{k+1}) - J_k(\theta, r_k)|. \quad (15)$$

The pair $(\beta_k, \mathcal{V}_K)$ separates approximation error in the frozen simulator from movement of the true objective sequence. Under the bounded-domain conditions below, both quantities are finite on every finite horizon; their asymptotic rates enter the online tracking guarantees in Section 5.

3.6 Projected residual and dynamic local regret

Fix a normalization floor $\vartheta > 0$ and a step size $\gamma > 0$. Define

$$\mathbf{d}(g) = \frac{g}{\max\{\|g\|_2, \vartheta\}}, \quad (16)$$

and the population projected residual

$$Q_k(\theta) = \frac{\Pi_\Theta\{\theta + \gamma \mathbf{d}(g_k(\theta))\} - \theta}{\gamma}. \quad (17)$$

The constrained first-order stationary set is

$$\mathbf{S}_k = \{\theta \in \Theta : Q_k(\theta) = 0\} = \{\theta \in \Theta : \langle g_k(\theta), u - \theta \rangle \leq 0 \ \forall u \in \Theta\}. \quad (18)$$

Because J_k is continuously differentiable on compact convex Θ , each J_k attains a maximizer satisfying (18); continuity of Q_k then makes $\mathbf{S}_k$ a nonempty compact set. At the realized iterate, let

$$\zeta_k := \|Q_k(\theta_k)\|_2^2, \quad \mathcal{E}_K := \frac{1}{K} \sum_{k=1}^K \mathbb{E} \zeta_k. \quad (19)$$

For window length $W \geq 1$ and discount $\rho \in (0, 1]$, define

$$A_{\rho, W} := \sum_{j=0}^{W-1} \rho^j, \quad \bar{Q}_k := A_{\rho, W}^{-1} \sum_{j=0}^{W-1} \rho^j Q_{k-j}(\theta_{k-j}), \quad (20)$$

with the convention $Q_t(\theta_t) = 0$ for $t \leq 0$. The dynamic local regret (DLR) is

$$\mathcal{R}_{\rho, W}(K) := \sum_{k=1}^K \|\bar{Q}_k\|_2^2. \quad (21)$$

Thus $\zeta_k = \|Q_k(\theta_k)\|_2^2$ records the periodwise first-order residual, while DLR first smooths the residual vectors over the recent window and then accumulates their squared norms.

3.7 Assumptions

The analysis rests on three standing conditions, each tied to a distinct part of the argument: uniform regularity of the portfolio trajectories, the source of policy-parameter dependence in the trajectory law, and smoothness of the regularized CPT weighting. More specialized conditions are introduced where they become relevant, so the main assumptions match the quantities used throughout the estimator and online analyses.

Assumption 3.1 (Compact policy and trajectory domain). *The parameter set $\Theta \subset \mathbb{R}^d$ is nonempty, compact, and convex, and is contained in an open set on which the policy is defined. The horizon h and asset count N are finite. There exist constants $Q_f, B_X, B_r, B_v < \infty$ such that, uniformly over every analyzed true model P_k and frozen simulator $\tilde{P}_k$,*

$$\sup_{i,u} \|q_{i,u}\|_2 \leq Q_f, \quad 0 < X_u \leq B_X, \quad 0 \leq r_u \leq B_r, \quad 0 \leq v_{\pm}(Y_k) \leq B_v. \quad (22)$$

Assumption 3.1 supplies a common domain on which score moments, objective values, and smoothness constants remain uniform across episodes. The parameter set, horizon, asset count, and feature norms are design-controlled; the wealth, reference, and value bounds describe the region on which the analysis is applied. The same fixed-target arguments can also be used on a local or high-probability bounded region, with the resulting constants interpreted on that region.

Assumption 3.2 (Policy-mediated likelihood structure). *Conditional on the observed history and executed action, the market transition kernel and the history-measurable feasibility map $\mathbf{M}_u(\cdot \mid s_u)$ have no explicit dependence on θ . The latent Dirichlet proposals $\tilde{\omega}_u$ are retained in the augmented trajectory, and each frozen simulator is held fixed as θ varies. Hence the policy density is the only explicit source of θ -dependence in the conditional augmented-trajectory law.*

This condition isolates the parameter dependence differentiated by the trajectory score in (8). It is satisfied when the market and feasibility mechanisms depend on the policy through the executed action and observed history, as in the simulator used here. Models with direct parameter-dependent transition dynamics require the corresponding likelihood or pathwise derivative term in the score identity.

Assumption 3.3 (Smooth regularized CPT weighting). *The preference parameters in (9)–(10) are fixed. The normalized weighting functions $w_{\pm}^{\varepsilon}$ are strictly increasing and belong to $C^3([0, 1])$, with*

$$C_j := \max_{\sigma \in \{+, -\}} \sup_{p \in [0, 1]} |(w_{\sigma}^{\varepsilon})^{(j)}(p)| < \infty, \quad j = 1, 2, 3. \quad (23)$$

Assumption 3.3 links the chosen regularization directly to the statistical construction. The first derivative enters the score-gradient identity, the second controls objective smoothness, and the third bounds the second-order remainder in the coupled full/half correction. Uniform endpoint regularity therefore provides one common set of constants for the gradient, debiasing, and online arguments.

Under Assumptions 3.1–3.3, Appendix A establishes finite uniform score moments and constants $B_g, L < \infty$ such that, for both true and frozen-simulator objectives,

$$\sup_{k, \theta \in \Theta} \|g_k(\theta)\|_2 \leq B_g, \quad \|g_k(\theta) - g_k(\theta')\|_2 \leq L\|\theta - \theta'\|_2. \quad (24)$$

In particular, the simulator-gradient discrepancy β_k and objective-variation budget $\mathcal{V}_K$ in (14)–(15) are finite for every finite experiment.

4 Algorithm

The learning procedure follows the decomposition introduced above. For a frozen episode context, the first stage constructs a gradient estimate whose expectation matches the regularized CPT gradient despite the nonlinear probability weighting. The second stage averages independent outputs and uses the resulting direction in a projected online update, while the realized portfolio path simultaneously updates wealth and the reference state.

4.1 Centered leave-one-out base and raw randomized correction

Fix one frozen episode context and parameter, and suppress the episode index. A complete trajectory yields

$$\mathbf{X}_i = (V_i^+, V_i^-, G_i), \quad V_i^\pm = v_\pm(Y_i). \quad (25)$$

For $\sigma \in \{+, -\}$, let $\varsigma_+ = 1$, $\varsigma_- = -1$ and define

$$I_{i,\sigma}(y) = \mathbf{1}\{V_i^\sigma > y\}, \quad \hat{S}_{-i,n}^\sigma(y) = \frac{1}{n-1} \sum_{j \neq i} I_{j,\sigma}(y). \quad (26)$$

The centered leave-one-out statistic is

$$U_n^{\text{cen}} = \frac{1}{n} \sum_{i=1}^n G_i \sum_{\sigma \in \{+,-\}} \varsigma_\sigma \int_0^{B_v} (w_\sigma^\varepsilon)'(\hat{S}_{-i,n}^\sigma(y)) \{I_{i,\sigma}(y) - \hat{S}_{-i,n}^\sigma(y)\} dy. \quad (27)$$

The corresponding raw statistic is

$$U_n^{\text{raw}} = \frac{1}{n} \sum_{i=1}^n G_i \sum_{\sigma \in \{+,-\}} \varsigma_\sigma \int_0^{B_v} (w_\sigma^\varepsilon)'(\hat{S}_{-i,n}^\sigma(y)) I_{i,\sigma}(y) dy. \quad (28)$$

Choose $n \geq 2$, activation probability $p_{\text{act}} \in (0, 1]$, and exponent $b \in (1, 2)$. Let

$$\nu_\ell = (1 - 2^{-b})2^{-b(\ell-1)}, \quad \ell = 1, 2, \dots \quad (29)$$

For $n_\ell = n2^\ell$, compute $U_{n_\ell}^{\text{raw}}$ on one batch of n_ℓ trajectories and the two half-sample statistics on its two disjoint halves. Define

$$D_{n,\ell}^{\text{raw}} = U_{n_\ell}^{\text{raw}} - \frac{1}{2} \left(U_{n_{\ell-1}}^{\text{raw},(1)} + U_{n_{\ell-1}}^{\text{raw},(2)} \right). \quad (30)$$

Let $A \sim \text{Bernoulli}(p_{\text{act}})$ and, conditional on $A = 1$, let $\ell^* \sim \nu$; both are independent of the base sample and the correction batch. One complete debiased output is

$$Z_n^{\text{db}} = U_n^{\text{cen}} + \frac{A}{p_{\text{act}}\nu_{\ell^*}} D_{n,\ell^*}^{\text{raw}}, \quad (31)$$

with the second term defined as zero when $A = 0$.

Algorithm 1 One centered-LOO debiased CPT-gradient output

Require: $(z, r, \tilde{P}, \theta)$, $n \geq 2$, $p_{\text{act}} \in (0, 1]$, $b \in (1, 2)$

- 1: $\mathcal{X}_0 \leftarrow \{\mathbf{X}_i\}_{i=1}^n$ i.i.d. from the frozen law at (z, r, θ) ; $U_0 \leftarrow U_n^{\text{cen}}(\mathcal{X}_0)$
- 2: $A \sim \text{Bernoulli}(p_{\text{act}})$
- 3: **if** $A = 0$ **then**
- 4: **return** U_0
- 5: **end if**
- 6: $\ell \sim \nu$; $\mathcal{X}_\ell \leftarrow \{\mathbf{X}_i\}_{i=1}^{n2^\ell}$ i.i.d. from the same frozen law, independently of $\mathcal{X}_0$
- 7: $D \leftarrow U_{n2^\ell}^{\text{raw}}(\mathcal{X}_\ell) - \frac{1}{2}\{U_{n2^{\ell-1}}^{\text{raw}}(\mathcal{X}_\ell^{(1)}) + U_{n2^{\ell-1}}^{\text{raw}}(\mathcal{X}_\ell^{(2)})\}$
- 8: **return** $Z_n^{\text{db}} = U_0 + D/(p_{\text{act}}\nu_\ell)$

The level distribution is untruncated: no deterministic cap is imposed on ℓ^* , and completed outputs are not discarded as a function of the realized level or magnitude.

4.2 Online update

The online loop uses the same stochastic Dirichlet actor defined in (6). Each episode first estimates the frozen-simulator gradient and updates the policy parameter after the realized five-step path has propagated the wealth, reference, and portfolio state.

In episode k , generate M_k conditionally independent copies of (31) under the frozen simulator and set

$$\hat{g}_k^{\text{db}} = \frac{1}{M_k} \sum_{m=1}^{M_k} Z_{k,m}^{\text{db}}. \quad (32)$$

The parameter update is

$$\theta_{k+1} = \Pi_\Theta \left\{ \theta_k + \gamma \mathbf{d}(\hat{g}_k^{\text{db}}) \right\}. \quad (33)$$

Algorithm 2 Dynamic-reference CPT policy gradient (DRCPT-PG)

Require: $\theta_1 \in \Theta$, (X_1, r_1, ω_0) , $h, \chi, \gamma, \vartheta$, (n, p_{act}, b) , $\{M_k\}_{k=1}^K$

- 1: **for** $k = 1, \dots, K$ **do**
- 2: fix $(z_k, r_k, \tilde{P}_k)$ conditionally on $\mathcal{F}_k$
- 3: generate $Z_{k,1}^{\text{db}}, \dots, Z_{k,M_k}^{\text{db}}$ independently by Algorithm 1
- 4: $\hat{g}_k^{\text{db}} \leftarrow M_k^{-1} \sum_{m=1}^{M_k} Z_{k,m}^{\text{db}}$
- 5: **for** $u = t_k, \dots, t_k + h - 1$ **do**
- 6: $a_{i,u} \leftarrow \exp(\theta_k^\top q_{i,u})$ for $i = 0, \dots, N$; $\tilde{\omega}_u \sim \text{Dirichlet}(a_{0,u}, \dots, a_{N,u})$
- 7: $\omega_u \leftarrow (1 - \chi)\omega_{u-1} + \chi M_u(\tilde{\omega}_u \mid s_u)$
- 8: $X_{u+1} \leftarrow X_u \{1 + \omega_u^\top R_{u+1} - c_{\text{tc}} \|\omega_u - \omega_{u-1}\|_1\}$; $r_{u+1} \leftarrow \Phi(r_u, X_{u+1})$
- 9: **end for**
- 10: $\theta_{k+1} \leftarrow \Pi_\Theta\{\theta_k + \gamma d(\hat{g}_k^{\text{db}})\}$
- 11: **end for**
- 12: **return** $\{\theta_k, X_{t_k}, r_k, \omega_{t_k-1}\}_{k=1}^{K+1}$

5 Main Results

The theory follows the same chain as the method. Endogenous reference updating first changes the objective itself; reliable optimization then requires a statistically controlled gradient for each frozen target and a procedure that can follow the resulting sequence of moving first-order conditions. We therefore establish the reference effect first, complete the frozen-target statistical theory next, and then turn to online tracking.

5.1 Reference dependence

Proposition 5.1 (Same terminal wealth, different CPT value). *Let $x_{\text{lo}} < r_0 < x_{\text{f}} < x_{\text{hi}}$ and consider*

$$\text{I} : r_0 \rightarrow x_{\text{hi}} \rightarrow x_{\text{f}}, \quad \text{II} : r_0 \rightarrow x_{\text{lo}} \rightarrow x_{\text{f}}.$$

Assume

$$x_{\text{f}} < (1 - \eta_+)r_0 + \eta_+x_{\text{hi}}, \quad x_{\text{f}} > (1 - \eta_-)r_0 + \eta_-x_{\text{lo}}.$$

Then

$$r_{\text{I}} - r_{\text{II}} = \eta_+(1 - \eta_-)x_{\text{hi}} - \eta_-(1 - \eta_+)x_{\text{lo}} + (\eta_- - \eta_+)x_{\text{f}}. \quad (34)$$

If the right-hand side is nonzero, the two paths have different regularized CPT values despite identical terminal wealth x_{f} .

Proposition 5.2 (Reference sensitivity). *Under Assumptions 3.1–3.3, fix (P, z, θ) . For this comparison only, suppose the policy features and market transition kernel are reference-isolated, so that changing the episode-start reference from r to r' while using common policy and market random variables leaves the wealth and portfolio paths unchanged. Set*

$$\eta_{\min} = \min(\eta_+, \eta_-), \quad L_v = \alpha_+ \varepsilon_v^{\alpha_+ - 1} + \lambda \alpha_- \varepsilon_v^{\alpha_- - 1}. \quad (35)$$

For two episode-start references r, r' coupled through the same wealth path,

$$|r_{t_k+h}(r) - r_{t_k+h}(r')| \leq (1 - \eta_{\min})^h |r - r'|, \quad (36)$$

and, for fixed (P, z, θ) ,

$$|J(P, z, r; \theta) - J(P, z, r'; \theta)| \leq C_1 L_v (1 - \eta_{\min})^h |r - r'|. \quad (37)$$

Consequently, with

$$\begin{aligned} \Delta_k^{\text{nr}} &:= \sup_{\theta \in \Theta} |J(P_{k+1}, z_{k+1}, r_k; \theta) - J(P_k, z_k, r_k; \theta)|, \\ \sup_{\theta \in \Theta} |J_{k+1}(\theta, r_{k+1}) - J_k(\theta, r_k)| &\leq \Delta_k^{\text{nr}} + C_1 L_v (1 - \eta_{\min})^h |r_{k+1} - r_k|. \end{aligned} \quad (38)$$

5.2 Exact frozen-target gradient estimation

Theorem 5.3 (Regularized CPT score-gradient identity). *Under Assumptions 3.1–3.3, fix one target and write $J = J_k$, $Y = Y_k$, $S_\theta^\pm(y) := S_{P_k, z_k, r_k, \theta}^\pm(y)$, and $G_\theta := G_{k, \theta}$. Define*

$$\begin{aligned} \Psi_\theta(\tau) &= \int_0^{B_v} (w_+^\varepsilon)'(S_\theta^+(y)) \{ \mathbf{1}[v_+(Y(\tau)) > y] - S_\theta^+(y) \} dy \\ &\quad - \int_0^{B_v} (w_-^\varepsilon)'(S_\theta^-(y)) \{ \mathbf{1}[v_-(Y(\tau)) > y] - S_\theta^-(y) \} dy. \end{aligned} \quad (39)$$

Then

$$\nabla J(\theta) = \mathbb{E}\{\Psi_\theta(\tau) G_\theta(\tau)\}. \quad (40)$$

The same identity holds for the frozen simulator objective $\tilde{J}$.

The empirical survival probabilities entering (39) make a direct finite-sample plug-in gradient biased. Appendix C shows that the leave-one-out base has $O(n^{-1})$ mean error and that the coupled full/half difference cancels the first-order influence term, leaving second moment $O(n^{-2}4^{-\ell})$. These two properties are the mechanism behind the randomized correction in Algorithm 1.

Theorem 5.4 (Exact unbiasedness, finite variance, and finite expected cost). *Let Z_n^{db} be defined by (31), with $n \geq 2$, $p_{\text{act}} \in (0, 1]$, and $1 < b < 2$. Let C_U and C_{rem} denote the finite constants established by Theorem C.1. Put*

$$\Xi_b := \sum_{\ell \geq 1} \frac{4^{-\ell}}{\nu_\ell} = \frac{1}{4(1 - 2^{-b})(1 - 2^{b-2})}, \quad \Upsilon_b := \sum_{\ell \geq 1} \nu_\ell 2^\ell = \frac{2(1 - 2^{-b})}{1 - 2^{1-b}}. \quad (41)$$

Then

$$\mathbb{E} Z_n^{\text{db}} = g, \quad (42)$$

$$\text{tr Cov}(Z_n^{\text{db}}) \leq C_{\text{var}}(n) := \frac{C_U}{n} + \frac{9C_{\text{rem}}\Xi_b}{p_{\text{act}}n^2} < \infty, \quad (43)$$

and the expected number of complete trajectories is

$$C_{\text{cost}}(n, p_{\text{act}}, b) = n(1 + p_{\text{act}}\Upsilon_b) < \infty. \quad (44)$$

For M independent outputs,

$$\hat{g}_{M,n}^{\text{db}} := \frac{1}{M} \sum_{m=1}^M Z_{n,m}^{\text{db}}, \quad \mathbb{E} \|\hat{g}_{M,n}^{\text{db}} - g\|_2^2 \leq \frac{C_{\text{var}}(n)}{M}. \quad (45)$$

Theorem 5.5 (Fixed-target asymptotic normality). *Under Theorem 5.4, fix (n, p_{act}, b) and let $\Sigma_n = \text{Cov}(Z_n^{\text{db}})$. Then*

$$\sqrt{M}(\hat{g}_{M,n}^{\text{db}} - g) \rightsquigarrow \mathcal{N}(0, \Sigma_n). \quad (46)$$

If Σ_n is nonsingular and $\hat{\Sigma}_{M,n}$ is the empirical covariance of the M outputs, then

$$\sqrt{M}\hat{\Sigma}_{M,n}^{-1/2}(\hat{g}_{M,n}^{\text{db}} - g) \rightsquigarrow \mathcal{N}(0, I_d). \quad (47)$$

For each frozen target, the untruncated construction is exactly unbiased with finite variance and finite expected trajectory cost. The CLT then provides replication-based uncertainty quantification for the same estimator.

5.3 Fixed-dimension minimax information rate

Definition 5.6 (Complete-trajectory oracle experiment). *Fix d , $\theta_0 = 0 \in \text{int}(\Theta)$, common finite bounds on h and N , and the constants in Assumptions 3.1–3.3. Let $\mathfrak{C}$ be a model class satisfying these bounds, containing the Bernoulli-return submodel constructed in Appendix E, with $(w_+^\varepsilon)'(0) > 0$. Under $P \in \mathfrak{C}$, an adaptive query $\theta_t \in \Theta$ returns one fresh complete augmented trajectory*

$$\tau_t \mid \mathcal{G}_{t-1} \sim P_{\theta_t}, \quad \mathcal{G}_t := \sigma\{(\theta_j, \tau_j) : j \leq t\}.$$

An admissible procedure chooses a stopping time T_{call} and an $\mathcal{G}_{T_{\text{call}}}$ -measurable estimator $\hat{g}$ such that

$$\sup_{P \in \mathfrak{C}} \mathbb{E}_P T_{\text{call}} \leq B, \quad (48)$$

with target $g_P := \nabla J_P(\theta_0)$.

The corresponding minimax risk is

$$\mathfrak{R}_B := \inf_{\hat{g}: \sup_{P \in \mathfrak{C}} \mathbb{E}_P T_{\text{call}} \leq B} \sup_{P \in \mathfrak{C}} \mathbb{E}_P \|\hat{g} - g_P\|_2^2. \quad (49)$$

Theorem 5.7 (Minimax-optimal fixed-dimension trajectory-budget rate). *For the oracle experiment in Definition 5.6, there exist constants $c_{\text{lb}}, C_{\text{ub}}, B_0 \in (0, \infty)$, independent of B , such that for all $B \geq B_0$,*

$$\frac{c_{\text{lb}}}{B} \leq \mathfrak{R}_B \leq \frac{C_{\text{ub}}}{B}. \quad (50)$$

The upper bound is attained, up to constants, by averaging independent untruncated outputs Z_n^{db} under the same expected complete-trajectory budget. Hence, for fixed d and fixed regularization, the minimax gradient-estimation rate is $\Theta(B^{-1})$.

5.4 Online first-order tracking under a moving CPT objective

Theorem 5.8 (Average expected squared projected residual bound). *Suppose Assumptions 3.1–3.3 hold and $\gamma L \leq \vartheta$. Let $\hat{g}_k^{\text{db}}$ be defined by (32) and target $\tilde{g}_k(\theta_k)$. Then*

$$\mathbb{E} \|\hat{g}_k^{\text{db}} - g_k(\theta_k)\|_2^2 \leq \frac{C_{\text{var}}(n)}{M_k} + \mathbb{E} \beta_k^2, \quad (51)$$

and

$$\boxed{\mathcal{E}_K \leq \frac{8\{2B_v + \mathbb{E} \mathcal{V}_K\}}{\vartheta \gamma K} + \frac{10C_g}{\vartheta^3 K} \sum_{k=1}^K \left(\frac{C_{\text{var}}(n)}{M_k} + \mathbb{E} \beta_k^2 \right)}, \quad (52)$$

Equation (52) separates the three quantities that govern online tracking: movement of the true objective sequence through $\mathcal{V}_K$, Monte Carlo error through M_k^{-1} , and frozen-simulator discrepancy through β_k^2 . The target is the contemporaneous constrained first-order condition rather than a single static optimum.

Corollary 5.9 (Dynamic local regret). *For every fixed window length $W \geq 1$, discount $\rho \in (0, 1]$, and horizon $K \geq 1$,*

$$\mathcal{R}_{\rho, W}(K) \leq \sum_{k=1}^K \zeta_k \quad (53)$$

pathwise, and hence

$$\frac{1}{K} \mathbb{E} \mathcal{R}_{\rho, W}(K) \leq \mathcal{E}_K. \quad (54)$$

Thus the same projected-residual control also governs the temporally smoothed first-order criterion used in the online evaluation.

Stationary-set interpretation. The projected residual is the primary first-order tracking quantity. A direct parameter-space interpretation follows under the additional local error-bound geometry: for some $\kappa < \infty$,

$$\text{dist}(\theta, \mathbf{S}_k) \leq \kappa \|Q_k(\theta)\|_2, \quad \theta \in \Theta, \quad (55)$$

uniformly over the analyzed episodes. Under this condition, residual control translates directly into distance from the contemporaneous constrained stationary set.

Corollary 5.10 (Moving stationary-set tracking). *If (55) holds, then*

$$\frac{1}{K} \sum_{k=1}^K \mathbb{E} \text{dist}^2(\theta_k, \mathbf{S}_k) \leq \kappa^2 \mathcal{E}_K. \quad (56)$$

Corollary 5.11 (Vanishing average expected squared projected residual and DLR). *Fix $W < \infty$ and $\rho \in (0, 1]$. If*

$$\mathbb{E} \mathcal{V}_K = o(K), \quad \frac{1}{K} \sum_{k=1}^K \left(\frac{1}{M_k} + \mathbb{E} \beta_k^2 \right) \rightarrow 0, \quad (57)$$

then

$$\mathcal{E}_K \rightarrow 0, \quad K^{-1} \mathbb{E} \mathcal{R}_{\rho, W}(K) \rightarrow 0. \quad (58)$$

If (55) holds, the left-hand side of (56) also converges to zero.

For the semi-synthetic environment used in E3–E4, Appendix D gives a sufficient objective-sensitivity condition under which the diminishing state gain and finite return-law interpolation imply sublinear objective variation. This links the theorem’s variation budget to the experimental construction.

6 Experimental Design and Evidence

6.1 Experimental roadmap

The empirical study proceeds in two layers. E1–E4 examine path-dependent reference formation, frozen-target gradient estimation, online first-order tracking, and ablation/robustness in controlled

settings. E5–E7 then evaluate chronological A-share performance, held-out investor behavior, and heterogeneous-agent interaction. Figure 2 summarizes the links among the seven studies.

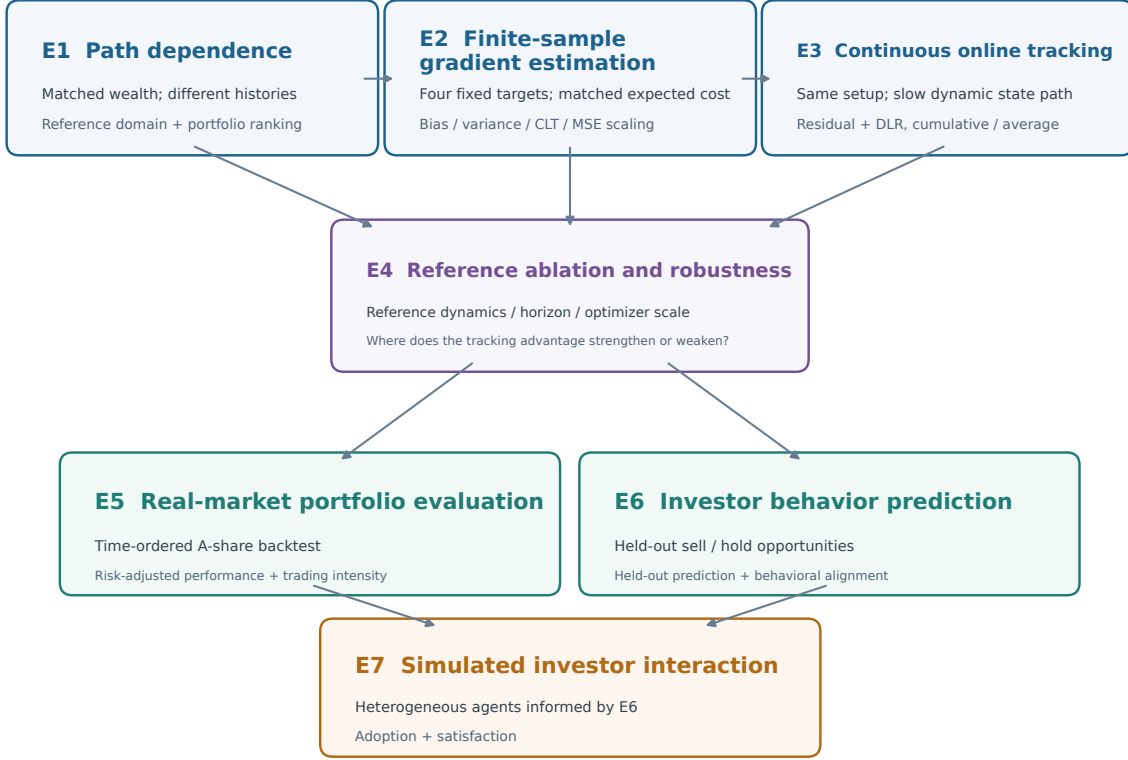


Figure 2: Empirical roadmap of the paper. E1–E4 establish path-dependent reference formation, finite-sample gradient estimation, online first-order tracking, and ablation/robustness in controlled settings. E5–E7 then extend the evaluation to chronological A-share performance, held-out investor behavior, and heterogeneous-agent interaction.

6.2 E1: Path dependence and portfolio choice

Design. We construct two histories with the same initial wealth and the same endpoint,

$$(1, 1.10, 1.015) \quad \text{and} \quad (1, 0.90, 1.015),$$

and apply the reference recursion in (3) with $(\eta_+, \eta_-) = (0.20, 0.05)$. The histories are matched at the decision time: current wealth is $X_T = 1.015$, the pre-trade portfolio is 50% cash and 50% risky asset, and the next risky return is +4% with probability p and −4% otherwise. We compare two feasible target allocations, 20% and 80% risky. Under the action inertia in (7), these targets execute as 44% and 56% risky. Because the next-period distribution has two states, all CPT values are evaluated exactly rather than by Monte Carlo.

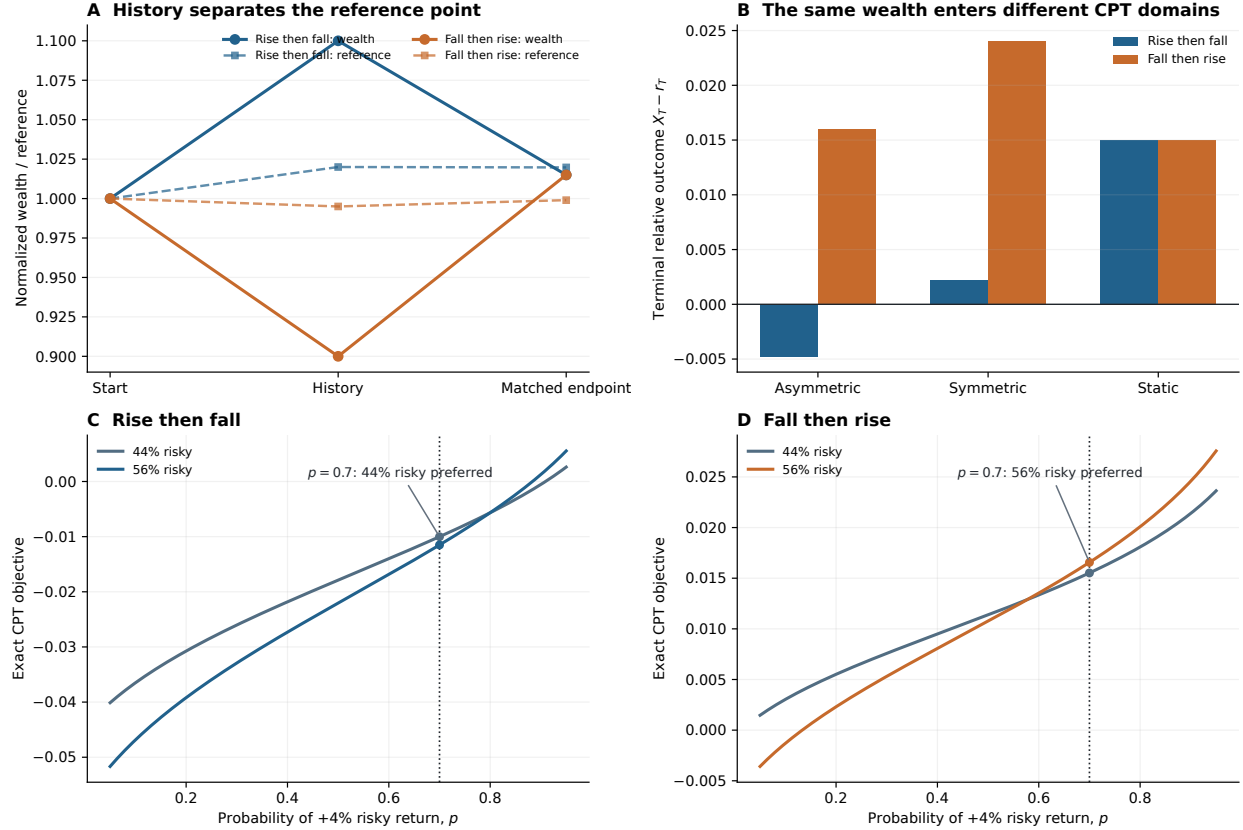


Figure 3: Path dependence changes both the perceived domain and the portfolio ranking. **A:** The two histories end at the same wealth but generate different asymmetric reference points. **B:** The asymmetric rule places the rise–fall history below its reference and the fall–rise history above its reference; the symmetric and static rules provide mechanism controls. **C–D:** Exact CPT values of the two feasible executed allocations as the probability p of a +4% risky return varies. At $p = 0.7$, the matched rise–fall account prefers 44% risky whereas the matched fall–rise account prefers 56% risky.

Table 1: E1 exact mechanism summary at $p = 0.7$.

History	X_T	r_T	$X_T - r_T$	Preferred risky share	CPT domain
Rise then fall	1.015	1.01975	−0.00475	44%	Loss
Fall then rise	1.015	0.99900	+0.01600	56%	Gain

Results. The key result is a rank reversal under matched current wealth, holdings, and investment opportunities. The two accounts have identical terminal wealth, pre-trade holdings, and next-period opportunity set; only their realized histories differ. Asymmetric adaptation places the rise–fall account below its reference, $X_T - r_T = -0.00475$, but the fall–rise account above its reference, $X_T - r_T = 0.016$. The same $\{+4\%, -4\%$ risky-return distribution is therefore evaluated on opposite branches of the CPT value function. At $p = 0.7$, this changes the preferred executed risky share from 44% after the rise–fall path to 56% after the fall–rise path. The change in ordering arises after current wealth, holdings, and opportunities have been held fixed, leaving the endogenous

reference as the state component that distinguishes the two histories. Dynamic reference formation is therefore decision-relevant: the same observable wealth and opportunity set can imply different feasible portfolio rankings under the behavioral criterion. This is the decision-level consequence of the path dependence characterized in [Theorem 5.1](#).

6.3 E2: Finite-sample gradient estimation

Shared semi-synthetic setup. Beginning with E2, we use a common 30-stock-plus-cash semi-synthetic market built from a frozen empirical A-share factor-block library. Factor blocks are sampled from a time-invariant empirical distribution, while the return law follows a pre-specified path across episodes. For risky asset i in episode k ,

$$R_{i,k,u} = 0.08 \tanh\left(\frac{\mu_k^{\text{ret}} + 2.5(\phi_k^{\text{ret}})^\top f_{i,u} + \xi_{k,u}^{\text{com}} + \xi_{i,k,u}^{\text{id}}}{0.08}\right),$$

where the common and idiosyncratic shocks have standard deviations $0.006\sigma_k^{\text{ret}}$ and $0.009\sigma_k^{\text{ret}}$, respectively; $(\mu_k^{\text{ret}}, \phi_k^{\text{ret}}, \sigma_k^{\text{ret}})$ denotes the episode-specific return-law coefficients. Episodes 1–40 use the stable momentum law, episodes 41–60 use the reversal law, episodes 61–120 interpolate to the terminal low-volatility/value/quality law, and the exogenous return law is fixed thereafter. E2 standardizes the account context to

$$(X, r, \omega, \theta) = (1, 1, e_{\text{cash}}, 0)$$

and freezes four pre-specified targets at episodes 20, 50, 90, and 150, corresponding to the stable, post-change, mid-drift, and post-drift phases. E3 subsequently uses the same factor distribution and episode-specific return law but allows the account state to evolve continuously.

We compare the proposed centered leave-one-out estimator with independent raw randomized correction against the raw independent-split plug-in estimator used by the adapted CPT-PG v5 baseline. Both target the square-smoothed CPT objective in (9). Expected complete-trajectory cost is matched over

$$B \in \{32, 64, 128, 256, 512, 1024, 2048\},$$

and numerical gradient references are generated independently at substantially higher cost.

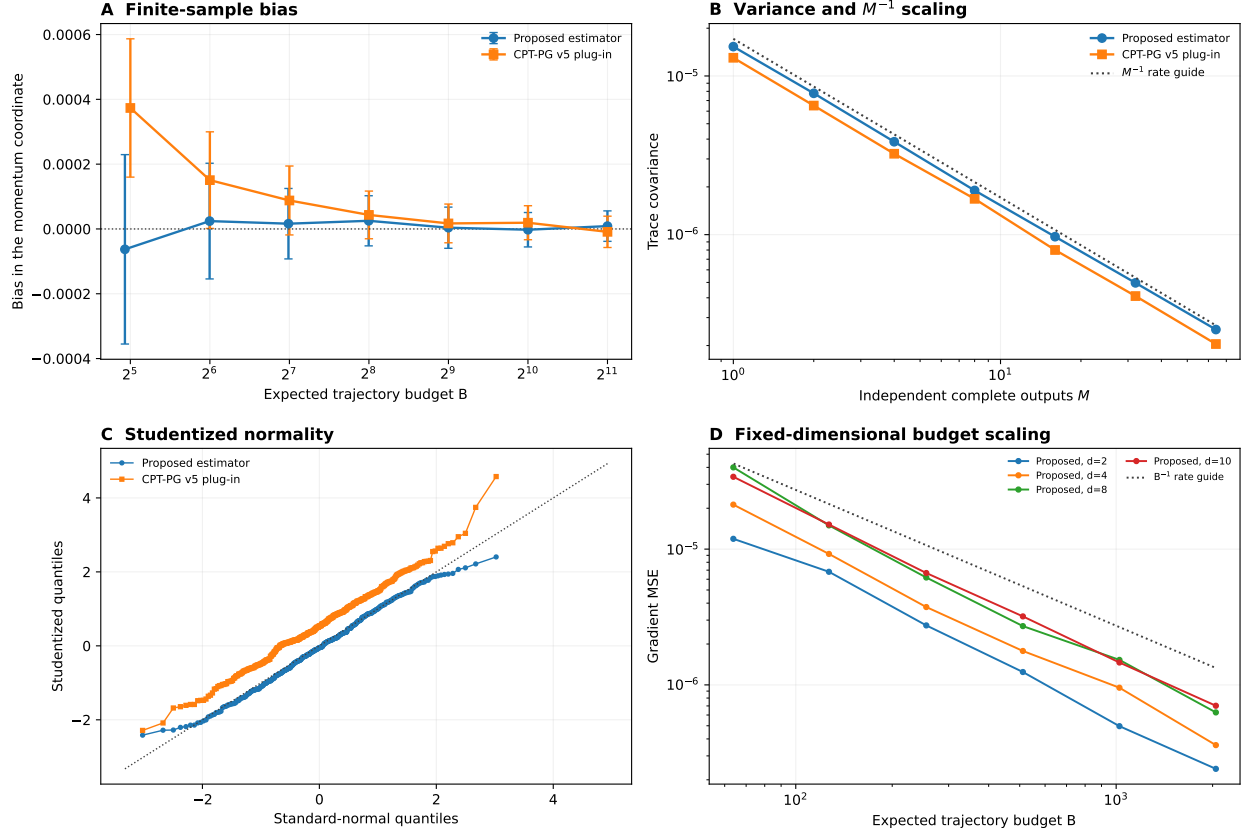


Figure 4: Finite-sample gradient estimation in the shared semi-synthetic setup. **A:** coordinate bias at the stable target. **B:** trace covariance of averages of M independent complete estimator outputs; the dotted reference has slope -1 . **C:** studentized normal-quantile diagnostic at $M = 64$. **D:** full-gradient MSE versus expected trajectory budget for $d \in \{2, 4, 8, 10\}$; the dotted reference has slope -1 . The corresponding unbiasedness, asymptotic normality, and fixed-dimensional minimax statements are given in [Theorems 5.4](#), [5.5](#) and [5.7](#).

Table 2: Full-gradient MSE at expected trajectory budget $B = 512$ across the four fixed targets. MSE entries are in units of 10^{-6} . The last column gives a paired 95% bootstrap interval for $\text{MSE}_{\text{prop}} - \text{MSE}_{\text{v5}}$ in the same units.

Target	Proposed	CPT-PG v5	Ratio	Difference 95% CI
Stable	3.123	2.938	1.063	$[-0.055, 0.434]$
Post-change	10.134	10.809	0.938	$[-1.495, 0.139]$
Mid-drift	10.647	11.468	0.928	$[-1.739, 0.102]$
Post-drift	10.589	12.257	0.864	$[-2.599, -0.721]$

Results. E2 separates three questions that an online gradient oracle must answer: centering, statistical efficiency, and inferential calibration. First, the proposed coordinate estimate stays centered near the numerical reference across the budget range, whereas the plug-in construction shows the low-budget displacement expected when nonlinear probability weighting is evaluated at an empirical distribution ([Figure 4A](#)). This is the finite-sample effect targeted by the leave-one-out

centering and independent randomized correction.

Second, the correction retains ordinary Monte Carlo efficiency. Averaging independent complete outputs gives fitted trace-covariance slopes of -0.989 for the proposed estimator and -0.999 for CPT-PG v5, and the proposed full-gradient MSE scales with budget at slopes between -1.17 and -1.12 across $d = 2, 4, 8, 10$ (Figure 4B,D). The correction therefore changes the estimator’s center while retaining the inverse-budget behavior predicted by Theorem 5.7. Third, the studentized statistic is close to its nominal Gaussian calibration: at $M = 64$ its mean is -0.049 , standard deviation 0.993 , and 95% coverage 96.25% , compared with mean 0.556 and coverage 91.75% for the v5 statistic (Figure 4C).

The four frozen targets show where this distinction matters for the full gradient. At the standard budget $B = 512$, the proposed estimator has the lower MSE point estimate in three targets, and after the completed drift its MSE is 13.6% lower with a paired interval entirely below zero (Table 2). Taken together, the evidence identifies the estimator’s role more precisely than a generic lower-variance claim: it removes a nonlinear finite-sample centering error while preserving the $1/B$ information rate, which is the property needed before the gradient is propagated through the online state dynamics.

6.4 E3: Continuous slow-variation online tracking

Design. E3 releases the market setup introduced in E2 as one continuous online process. No account state is reset. Let $(X_k^{\text{raw}}, r_k^{\text{raw}}, \omega_k^{\text{raw}})$ denote the raw terminal account state produced by episode k before the cross-episode relaxation. The next episode starts from

$$\begin{aligned}\log X_{t_{k+1}} &= (1 - \varpi_k) \log X_{t_k} + \varpi_k \log X_k^{\text{raw}}, \\ r_{k+1} &= (1 - \varpi_k) r_k + \varpi_k r_k^{\text{raw}}, \\ \omega_{t_{k+1}-1} &= (1 - \varpi_k) \omega_{t_k-1} + \varpi_k \omega_k^{\text{raw}}, \quad \varpi_k = k^{-3/4}.\end{aligned}$$

The within-episode reference update remains the asymmetric mechanism in (3) with $(\eta_+, \eta_-) = (0.20, 0.05)$. Under the bounded analysis domain of Assumption 3.1 and the bounded semi-synthetic returns, the episode-start state increments are $O(\varpi_k)$, with

$$\sum_{k \leq K} \varpi_k = O(K^{1/4}) = o(K).$$

The factor-block distribution is time invariant, the exogenous return coefficients follow the E2 phase path through episode 120, and remain fixed for episodes 121–300. Appendix D shows how this schedule induces $\mathcal{V}_K = O(K^{1/4})$ whenever the frozen objective is uniformly Lipschitz in the episode-start account variables and return-law coefficients on the analyzed compact domain. E3 therefore controls the design-side quantities that enter the theorem’s variation budget while retaining the dynamic reference’s path dependence.

The comparison uses 20 independent test trajectories. The online learner and frozen control share the complete path through episode 40. Thereafter the online learner continues Algorithm 2, whereas the control freezes θ but continues to propagate wealth, reference, and holdings under the same slow transition. We fix $(\gamma, \vartheta) = (0.08, 0.05)$ before the E3 test runs and use the growing replication schedule

$$M_k = \max\{4, \lceil \sqrt{k} \rceil\},$$

for which $M_k \geq \sqrt{k}$ and $K^{-1} \sum_{k=1}^K M_k^{-1} \leq (2\sqrt{K} - 1)/K = O(K^{-1/2})$. DLR follows (21) with $W = 5$ and $\rho = 0.9$. Two independent evaluation streams form cross-products of the projected-residual estimates, both periodwise and after the same exponential vector smoothing used in

(20). Independence removes the ordinary self-square noise term, while (87) and (88) bound the remaining discrepancies from the population periodwise residual and DLR terms in terms of the evaluation-gradient MSE.

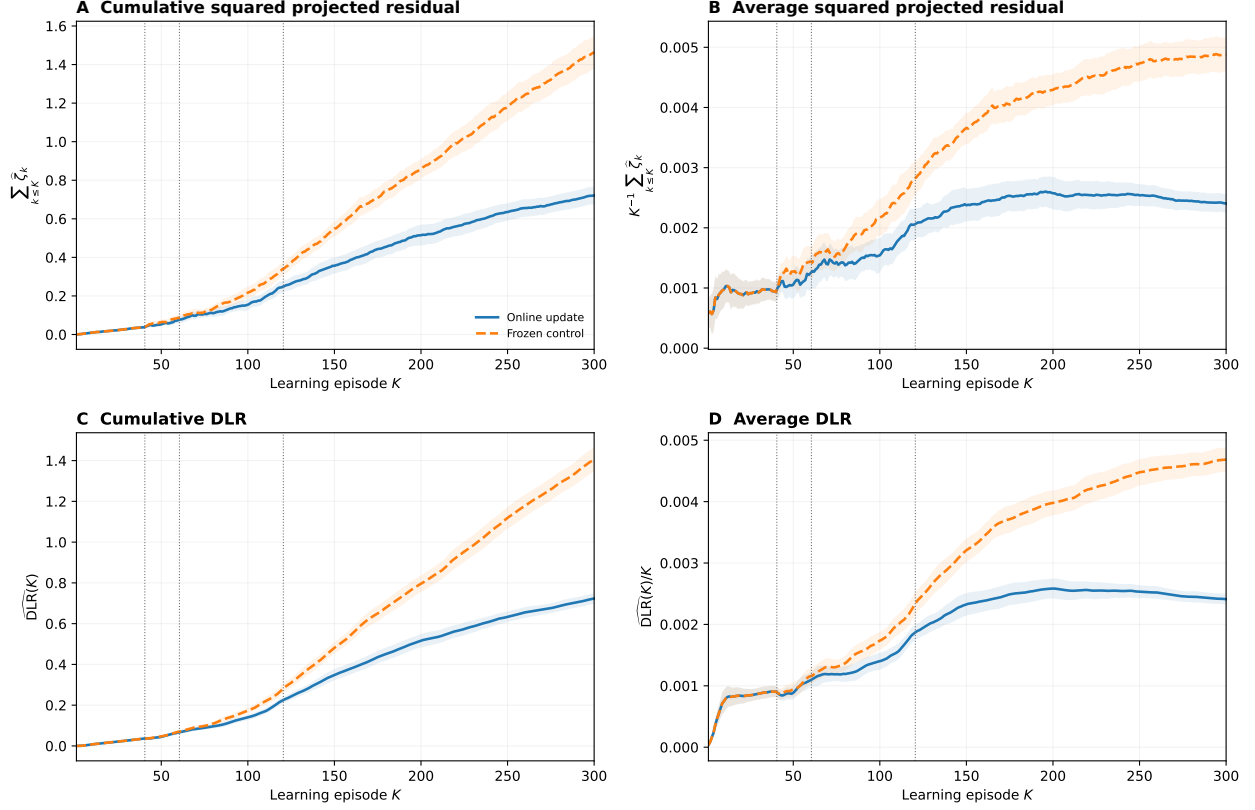


Figure 5: Continuous online tracking under slow variation. **A–B**: cumulative squared projected residual and its average. **C–D**: cumulative and average DLR. The two methods share the same trajectory through episode 40; thereafter the frozen control holds θ fixed while both methods continue to propagate their own wealth, reference, and holdings. Dotted lines mark the abrupt change, the start of gradual drift, and the point after which the exogenous return law is fixed. Curves are means over 20 independent trajectories and shaded bands are 1.96 standard errors.

Results. Continued adaptation changes the accumulation of first-order tracking error over the evolving portfolio path. From episodes 41–300, the online learner accumulates 0.684 squared projected residual versus 1.427 for the frozen control, a reduction of about 52%; cumulative DLR is 0.687 versus 1.369, a reduction of about 50%. The corresponding seed-paired 95% bootstrap intervals for the differences are $[-0.834, -0.650]$ and $[-0.738, -0.622]$, and by episode 300 the average DLR is 2.411×10^{-3} online versus 4.685×10^{-3} frozen (Figure 5). After episode 40, the two arms evolve through their own endogenous wealth, reference, and holding paths. The projected residual and DLR therefore measure how closely each induced closed-loop system follows its contemporaneous first-order condition.

The tail behavior strengthens this interpretation. After the exogenous return law becomes fixed, episodes 121–300 still contain slow endogenous state variation. Over this interval, cumulative DLR grows with empirical log–log slope 0.704 under continued updating but 0.968 under freezing; squared projected residual shows the same separation, 0.745 versus 0.959. Thus updating changes error

accumulation from a nearly linear frozen trajectory toward the sublinear direction characterized by [Theorems 5.9](#) and [5.11](#). The two independent evaluation streams remove the ordinary self-square noise term, and the residual-map bounds in (87)–(88) control the remaining nonlinear evaluation error through the gradient-estimation MSE. E3 therefore shows that repeated first-order corrections track the slowly moving target more closely over time. Under the local geometric condition (55), the same residual control also translates into average squared distance to the contemporaneous stationary set through [Theorem 5.10](#).

6.5 E4: Reference-mechanism ablation and robustness of online tracking

Design. E4 first asks how the reference-update mechanism changes the E3 tracking advantage, then tests whether the same advantage persists across decision horizons and optimizer scales. Because each perturbation changes the induced objective sequence, absolute DLR values are not compared across configurations. For configuration c and paired seed s , we form the matched-control ratio

$$\text{Ratio}_{\text{DLR}}(c, s) = \frac{\text{DLR}_{\text{online}}^{\text{post}}(c, s)}{\text{DLR}_{\text{frozen}}^{\text{post}}(c, s)},$$

where the post-change interval begins at the first regime change. Figures show the mean of the seed-level ratios with paired-seed bootstrap intervals, rather than the ratio of two across-seed mean DLR values. A ratio below one means that continued updating accumulates less DLR than freezing θ on that matched trajectory.

Panel A is a structured reference-mechanism ablation over the full 9×9 grid

$$\eta_+, \eta_- \in \{0, 0.05, 0.10, \dots, 0.40\}.$$

Panel B varies the episode horizon over

$$h \in \{1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50\},$$

with the 200-, 300-, 600-, and 1500-day boundaries assigned to the nearest complete episode where needed; the state-gain clock remains tied to elapsed market time. Panel C evaluates 48 optimizer configurations from

$$\gamma \in \{0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.08, 0.10, 0.12\},$$

$$\vartheta \in \{0.005, 0.01, 0.015, 0.02, 0.03, 0.04, 0.05, 0.06\}, \quad \gamma c_{\text{scr}} \leq \vartheta, \quad c_{\text{scr}} = 0.31.$$

The retained grid satisfies $\gamma c_{\text{scr}} \leq \vartheta$ with $c_{\text{scr}} = 0.31$. The E3 defaults $(\eta_+, \eta_-) = (0.20, 0.05)$, $h = 5$, and $(\gamma, \vartheta) = (0.08, 0.05)$ are marked throughout the robustness plots. Each configuration uses four independent paired trajectories and two independent evaluation streams.

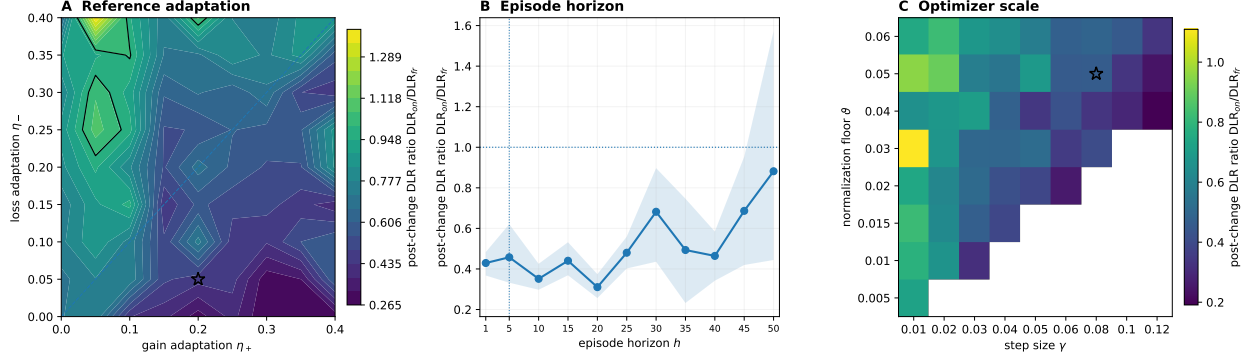


Figure 6: Reference-mechanism ablation and robustness of the online-tracking advantage, measured by the configuration-matched post-change DLR ratio $\text{Ratio}_{\text{DLR}}$. **A**: structured ablation over the 9×9 reference-update grid; $(0, 0)$ removes reference adaptation, the diagonal $\eta_+ = \eta_-$ removes gain–loss asymmetry while retaining adaptation, and the off-diagonal regions test asymmetric updating. The black contour marks a ratio of one and the star marks the default rule. **B**: horizon sweep with paired-seed 95% bootstrap intervals; the horizontal line marks a ratio of one and the vertical line marks $h = 5$. **C**: (γ, ϑ) robustness grid subject to $\gamma c_{\text{scr}} \leq \vartheta$; blank cells lie outside this region and the star marks the E3 default.

Results. Panel A isolates how the reference mechanism changes the tracking advantage. Removing reference adaptation altogether, $(\eta_+, \eta_-) = (0, 0)$, does not eliminate the benefit of online updating: the matched DLR ratio remains 0.716. Likewise, all eight positive symmetric-adaptation settings on the diagonal, $\eta_+ = \eta_- > 0$, remain below one, with a mean ratio of 0.715 and six paired 95% intervals entirely below one. Online updating therefore remains effective under static and symmetric references, while asymmetry changes the strength of the tracking advantage. Across all 36 gain-dominant configurations with $\eta_+ > \eta_-$, the mean ratio falls to 0.511, every configuration remains below one, and 33 intervals lie entirely below one. The loss-dominant region is weaker on average, with mean ratio 0.826, and sufficiently strong reversal of the asymmetry can overturn the comparison: at $(\eta_+, \eta_-) = (0.05, 0.40)$ the ratio is 1.403 with interval $[1.224, 1.552]$ (Figure 6A). Panel A thus identifies gain-dominant asymmetric reference adaptation as an amplifying and structuring mechanism for online tracking.

Panels B and C examine temporal and optimizer robustness. Mean ratios are below one at all 11 horizons, with 10 intervals entirely below one; only the $h = 50$ interval crosses one (Figure 6B). Across the 48 screened (γ, ϑ) pairs, 47 mean ratios are below one and 45 intervals are entirely below one; the lone point estimate above one has interval $[0.876, 1.342]$. Together, the panels show that the tracking advantage persists across a broad range of decision horizons and optimizer scales, while the reference ablation identifies where the strength of that advantage changes.

6.6 E5: Real-market portfolio evaluation

Design. E5 transfers the portfolio learner from the controlled E2–E4 market to a chronological A-share replay. The universe is fixed before the test period at 300 HS300 constituents. Historical inputs begin on January 3, 2023, and the out-of-sample execution window runs from January 2, 2025 through May 29, 2026. Every factor is point-in-time: momentum, reversal, volatility, valuation, liquidity, size, and previous holdings use only information available before the execution date, and accounting-quality variables enter only after their announcement dates. The learner and the realized

account use the same stochastic Dirichlet actor, and transaction cost $0.001\|\omega_t - \omega_{t-1}\|_1$ is deducted from wealth.

We compare five policy-gradient methods: DRCPT-PG, symmetric-reference CPT-PG, static-reference CPT-PG, expected-wealth PG, and exponential-utility PG. All methods use the same five-day horizon, $\gamma = 0.08$, expected trajectory budget 512, and ten pre-specified policy seeds; DRCPT-PG keeps $(\eta_+, \eta_-) = (0.20, 0.05)$. We evaluate market performance using wealth, annualized return and volatility, Sharpe ratio, maximum drawdown, turnover, and cumulative transaction cost.

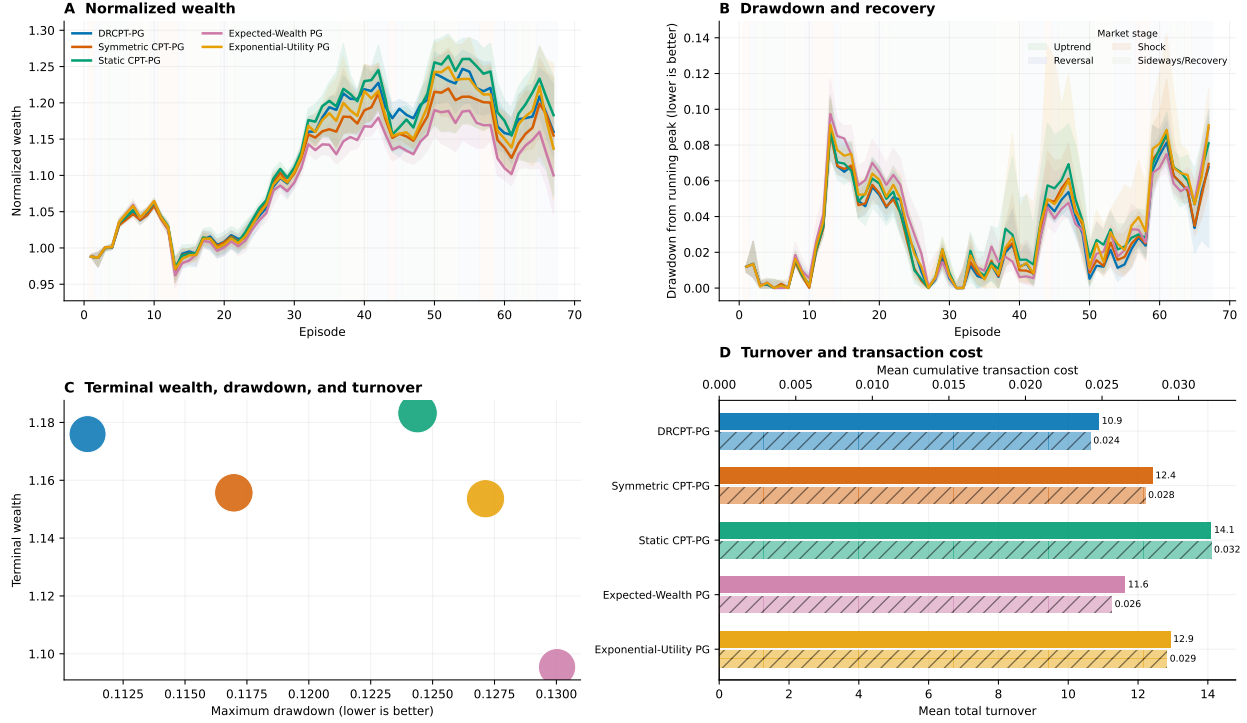


Figure 7: E5 chronological A-share evaluation. **A**: median normalized wealth across the ten policy seeds, with seed dispersion bands. **B**: drawdown and recovery over the same replay, with market-stage annotations indicating the main market phases. **C**: terminal wealth versus maximum drawdown; bubble size represents turnover. **D**: mean turnover and cumulative transaction cost. All five methods share the same fixed asset universe, point-in-time information set, transaction-cost rule, and stochastic execution protocol.

Table 3: E5 real-market summary. Entries are means across the ten policy seeds.

Method	Ann. return (%)	Sharpe	Max DD (%)	Turnover	Cost (%)
DRCPT-PG	13.03	0.856	11.67	10.87	2.42
Symmetric CPT-PG	13.21	0.796	12.48	12.41	2.78
Static CPT-PG	14.69	0.789	14.36	14.09	3.21
Expected-Wealth PG	10.51	0.606	14.23	11.63	2.56
Exponential-Utility PG	14.79	0.773	13.95	12.92	2.92

Results. E5 asks whether the dynamic-reference mechanism remains economically meaningful once the controlled return law is replaced by chronological A-share data. Across the five policy-gradient

methods, DRCPT-PG has the highest mean Sharpe ratio, 0.856, together with the lowest mean maximum drawdown, 11.67%, turnover, 10.87, and cumulative transaction cost, 2.42% (Table 3). The closest mechanism control is symmetric CPT-PG: at similar mean annualized return (13.03% versus 13.21%), DRCPT-PG lowers volatility by 4.5%, maximum drawdown by 6.5%, turnover by 12.4%, and transaction cost by 12.9%, increasing the Sharpe ratio by 7.6%. The separation therefore appears jointly in risk exposure and trading intensity rather than in a single endpoint. Combined with the static-reference comparison in Figure 7, this supports the market-level implication of the model: endogenous asymmetric reference adaptation changes how the policy allocates risk and how aggressively it trades under the same information set and execution protocol.

6.7 E6: Investor behavior prediction

Design. E6 asks whether the reference state used by the portfolio model carries incremental information about real investor decisions. The sample contains 306,444 held-stock rebalance opportunities, with 71,854 observations in the final chronological test split. The outcome is whether the investor decreases the stock’s target weight at the next rebalance. Four logistic specifications use exactly the same observations, outcome, controls, and train–validation–test dates; they differ only in the reference representation: no reference, static purchase cost, symmetric adaptation, or asymmetric adaptation with $(\eta_+, \eta_-) = (0.40, 0.10)$. Common controls include holding age, previous weight, price, prior investor reduction behavior, and lagged market, portfolio, and social aggregates. Uncertainty is assessed by 200 paired investor-cluster bootstrap draws.

The behavioral statistic is recomputed under each reference representation rather than fixing gain and loss states to the static purchase cost. For the same realized decisions, we therefore report the reduction rate when the position is above its own reference, the reduction rate when it is below its own reference, and their difference. This separates two questions: whether a reference representation improves held-out prediction and whether the gain/loss partition that it induces aligns with the observed disposition asymmetry.

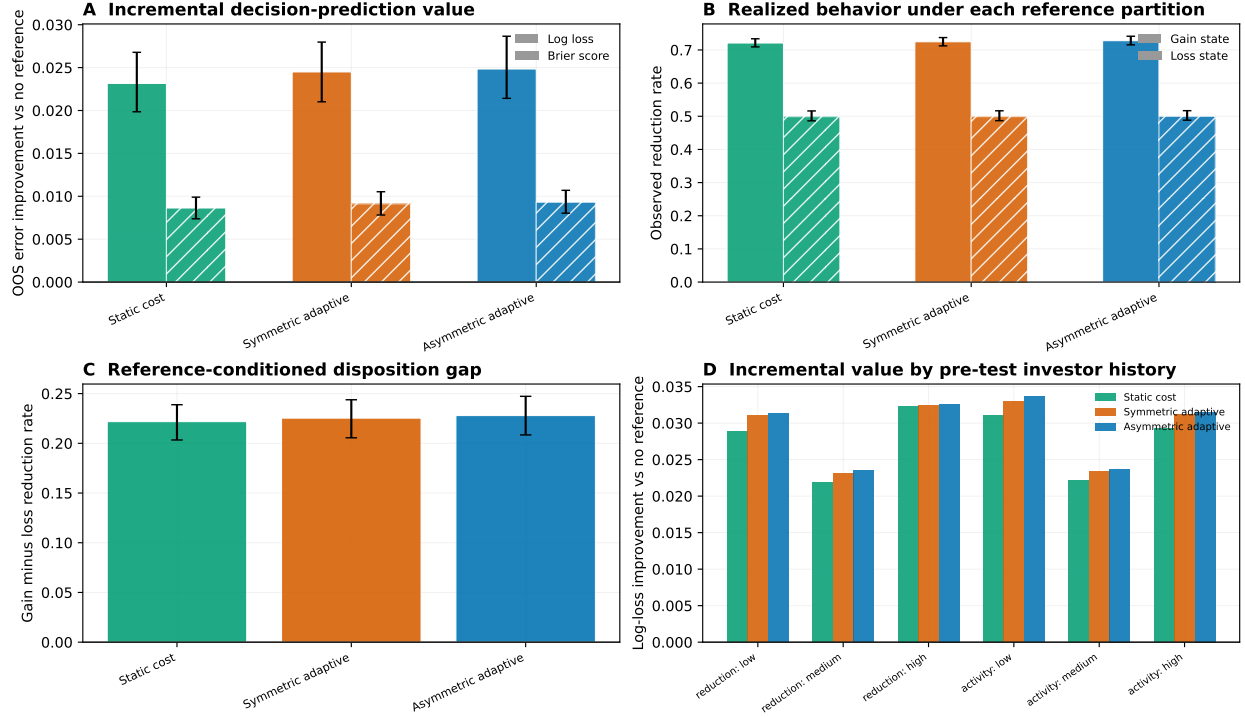


Figure 8: E6 held-out behavioral alignment. **A**: paired-bootstrap improvements in out-of-sample log loss and Brier score relative to the no-reference model. **B**: observed reduction rates in gain and loss states defined separately by each reference. **C**: reference-conditioned gain-minus-loss reduction gap. **D**: log-loss improvement over the no-reference model across pre-test investor-history strata. Error bars in A–C are paired investor-cluster 95% bootstrap intervals.

Results. E6 provides two complementary tests of behavioral alignment: out-of-sample prediction and the gain–loss partition induced by the reference itself. Introducing reference information produces a clear predictive gain, and the improvement continues as the representation becomes more behaviorally structured. On the 71,854 held-out decisions, log loss falls from 0.60420 with no reference to 0.58109 with a static reference, 0.57975 with symmetric adaptation, and 0.57937 with asymmetric adaptation. The asymmetric-minus-none paired-bootstrap difference is -0.02488 with 95% interval $[-0.02866, -0.02142]$; Brier score, AUROC, and AUPRC move in the same direction (Figure 8A). Relative to the static reference, asymmetric adaptation further reduces log loss by 0.00174 with paired interval $[-0.00236, -0.00123]$, while the asymmetric-minus-symmetric difference remains resolved at -0.000391 with interval $[-0.000695, -0.000014]$. Thus the data distinguish three nested components: representing a reference state, allowing that state to adapt, and allowing gain–loss asymmetry in the adaptation.

The behavioral partition tells the same story. The gain-minus-loss reduction gap rises from 0.2214 under the static reference to 0.2250 under symmetric adaptation and 0.2274 under asymmetric adaptation. The paired-bootstrap asymmetric gap is 0.2279 with interval $[0.2085, 0.2473]$, and it exceeds the symmetric gap by 0.00255 with interval $[0.00107, 0.00404]$ (Figure 8B,C). The asymmetric specification also gives the largest log-loss improvement in all six pre-test history strata (Figure 8D). Prediction and reference-conditioned behavior therefore point to the same conclusion: the asymmetric adaptive-reference representation carries incremental information about held-out investor decisions.

6.8 E7: Simulated heterogeneous-investor interaction

Design. E7 adds an interaction layer without retraining the portfolio policies. We calibrate four investor types from 3,481 real users using training-period E6 gain/loss reduction statistics together with historical activity, diversification, sentiment, and network summaries. The resulting population shares are 49.4% moderate-frequency focused investors, 28.7% disposition-effect investors, 19.4% active diversified investors, and 2.5% low-frequency diversified investors. Sixteen calibrated users are sampled per type, producing 64 simulated agents. Type construction uses training-period behavioral summaries only; the global E6 predictive coefficients are not clustering features.

For each of the five policy-gradient methods, the experiment exactly replays the full cash-plus-300-stock recommendations saved by E5. At the beginning of each five-day episode, adoption is sampled from the current recommendation, calibrated investor history, compatibility, and lagged satisfaction before any return from the episode is observed. If the recommendation is adopted, the agent executes that method for five days; otherwise it keeps the episode-start portfolio. Both paths apply their own trading costs and reference updates. Conditional satisfaction is the five-day normalized prospect-value improvement over the no-rebalance counterfactual, and interaction value multiplies that satisfaction by adoption so that rejected recommendations contribute zero. Seed-level market paths and adoption draws are paired across methods, and population summaries are weighted by the empirical type shares.

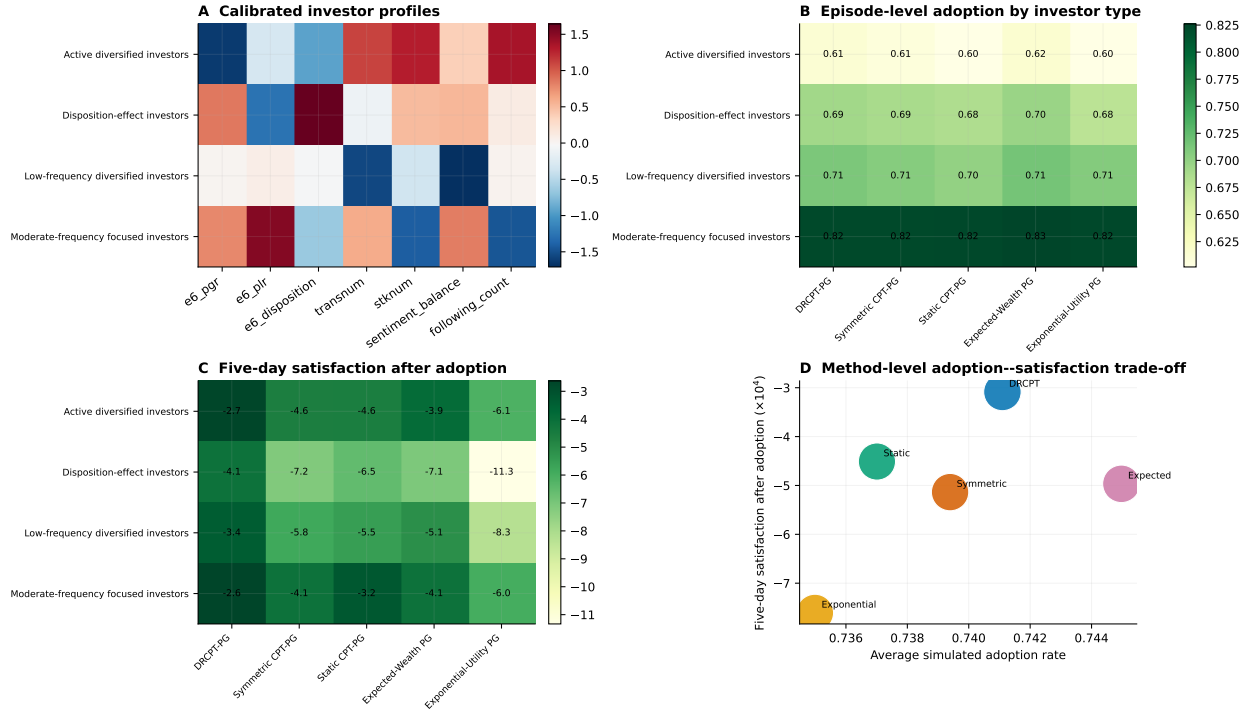


Figure 9: E7 calibrated heterogeneous-agent interaction. **A**: standardized profiles of the four investor types. **B**: simulated episode-level adoption by type and method. **C**: five-day conditional satisfaction after adoption. **D**: method-level adoption-satisfaction plane; bubble size encodes mean recommendation compatibility. Adoption is decided before episode returns are revealed, and satisfaction is evaluated against a no-rebalance counterfactual with its own reference path.

Results. E7 separates two quantities that would be conflated by an adoption-only evaluation. Population-weighted adoption rates are tightly clustered across the five advisers, from 0.735 to 0.745, so uptake alone cannot explain the method ordering. DRCPT-PG adopts at rate 0.741, but its conditional five-day satisfaction, -3.087×10^{-4} , and adoption-weighted interaction value, -2.265×10^{-4} , are both the most favorable among the five methods. Relative to static, symmetric, expected-wealth, and exponential-utility PG, the magnitude of the interaction-value shortfall is smaller by approximately 30%, 39%, 38%, and 59%, respectively. Expected-wealth PG, for example, has a slightly higher adoption rate (0.745) but a lower post-adoption value, which is why the adoption–satisfaction plane in Figure 9D is more informative than uptake alone.

The ordering persists across all four calibrated investor types: DRCPT-PG has the highest adoption-weighted interaction value for active diversified, disposition-effect, low-frequency diversified, and moderate-frequency focused investors, while also using the lowest mean turnover (9.29) and cumulative transaction cost (0.0208) among the five advisers. The calibrated replay therefore distinguishes the advisers primarily by post-adoption quality rather than uptake. DRCPT-PG gives the most favorable interaction value across all four investor types, while the negative population means relative to the no-rebalance benchmark leave absolute welfare improvement as a separate question for prospective evaluation. This interaction-level result complements E6 by showing that the dynamic-reference advantage persists across markedly different calibrated investor profiles.

7 Discussion

Endogenous reference formation changes a CPT portfolio problem at two levels at once. It changes the economic state being optimized, because identical current wealth can be evaluated differently after different gain–loss histories, and it changes the learning problem, because the resulting objective moves with both market conditions and the investor’s own realized path. The results connect these two levels within one first-order framework: a frozen dynamic-reference target admits statistically controlled gradient estimation, and the online update can track the sequence of moving first-order conditions through the projected residual and dynamic local regret.

This connection extends two strands of prior work that have largely been treated separately. Fixed-reference CPT reinforcement learning establishes how nonlinear probability weighting can be optimized with policy gradients [3, 4], while adaptive-reference models in behavioral finance show that realized gains and losses can shift the benchmark against which subsequent outcomes are judged [5–8]. DRCPT-PG combines these ideas by making the reference an endogenous optimization state. E1 makes the consequence concrete at the decision level, and E3–E4 show that once the target moves with that state, continued first-order updating reduces the accumulation of tracking error across a broad range of reference dynamics, horizons, and optimizer scales.

The market and behavioral studies indicate that the same mechanism remains informative beyond the controlled construction. In the chronological A-share replay, DRCPT-PG achieves the highest mean Sharpe ratio and the lowest mean maximum drawdown, turnover, and transaction cost among the five policy-gradient methods. The pattern is economically coherent with the model: relative to fixed or symmetric references, asymmetric reference adaptation changes both risk exposure and trading intensity under the same point-in-time information and execution protocol. Held-out investor data provide a complementary behavioral test. Reference-based specifications improve out-of-sample prediction, and adaptive and asymmetric representations add further predictive information while sharpening the gain–loss partition associated with observed reductions. The calibrated heterogeneous-agent replay then extends this comparison to repeated recommendation interactions, where adoption rates are similar across advisers but post-adoption interaction values

favor DRCPT-PG across all four investor types.

Taken together, the evidence suggests a division of roles that is useful beyond this particular implementation. The reference recursion determines how realized experience reshapes the behavioral objective; the gradient estimator supplies a reliable local first-order signal for that objective; and the online update determines how closely the policy follows the resulting moving target. This view helps explain why the reference mechanism and online adaptation are complementary rather than interchangeable. E4 shows that online updating remains useful under static and symmetric references, while gain-dominant asymmetry strengthens the tracking advantage. The behavioral state therefore structures the target being followed, and the optimizer governs how effectively the learner follows it.

Several open questions follow from this framework. Richer debiasing schemes may reduce the trajectory cost of the frozen-target oracle while retaining exactness, and sharper objective-sensitivity conditions may extend the online guarantees to more abrupt market and reference shifts. On the behavioral side, a prospective human-in-the-loop study could jointly observe reference adaptation, recommendation uptake, realized portfolio outcomes, and reported utility under a common protocol, providing a direct test of how the modeled reference state maps to experienced investor preferences. Taken together, the framework turns adaptive reference formation from a behavioral specification into an optimization state that can be estimated, tracked, and evaluated within a unified first-order portfolio-learning framework.

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A Notation and Analytic Preliminaries

All fixed-target statistical statements condition on the episode-start information, the queried parameter, and the frozen simulator before new Monte Carlo trajectories are drawn. Conditional independence is always with respect to this law. Limits in M fix the target and estimator design (n, p_{act}, b) ; limits in K require the constants in the stated assumptions to be uniform over episodes; the minimax limit fixes d and all model-class constants while $B \rightarrow \infty$.

A.1 Main notation

Table 4: Main notation along the state, objective, estimation, and online-evaluation chain. Estimator internals and proof-local constants are defined where they are used.

Symbol	Type	Meaning
k, u, h	indices / integer	episode, calendar time, and finite episode horizon
$s_u, \omega_u, \tilde{\omega}_u$	state / portfolios	state $(m_u, X_u, r_u, \omega_{u-1})$ with market state m_u ; executed portfolio and latent proposal
$z_k, \mathcal{F}_k, r_k$	context / information	frozen non-reference start state, episode-start information, and reference r_{t_k}
$\theta, \Theta, \pi_\theta$	vector / set / policy	parameter, compact convex parameter set, and Dirichlet actor
$P_k, \tilde{P}_k$	episode models	true and frozen-simulator models inducing policy-dependent trajectory laws
$Y_k, v_\pm, w_\pm^\varepsilon$	outcome / functions	terminal relative outcome and regularized CPT value and weight maps
$J_k, \tilde{J}_k$	scalar functions	true and frozen-simulator CPT objectives
$g_k, \tilde{g}_k, G_{k,\theta}$	$\mathbb{R}^d$	true gradient, simulator gradient, and complete-trajectory score
$\hat{g}_k^{\text{db}}, M_k$	vector / integer	debiased update estimate averaged over M_k independent complete outputs
$Q_k; \gamma, \vartheta$	vector / scalars	projected residual; step size and normalization floor
S_k	subset of Θ	constrained first-order stationary set
ζ_k	$\mathbb{R}_+$	squared projected residual $\ Q_k(\theta_k)\ _2^2$
$\mathcal{E}_K$	$\mathbb{R}_+$	average expected squared projected residual
$A_{\rho, W}, \bar{Q}_k$	scalar / $\mathbb{R}^d$	full-window exponential mass and temporally smoothed projected residual
$\mathcal{R}_{\rho, W}(K); \rho, W$	scalar / parameters	dynamic local regret; discount and window length
$\beta_k, \mathcal{V}_K$	$\mathbb{R}_+$	simulator-gradient error bound and cumulative true-objective variation
T_{call}, B	integer / scalar	oracle trajectory count and expected trajectory budget

A.2 Dirichlet score moments

Lemma A.1 (Uniform moments of the augmented trajectory score). *Under Assumptions 3.1–3.2, there exist $0 < \underline{a} \leq \bar{a} < \infty$ such that*

$$\underline{a} \leq a_{i,u}(\theta) \leq \bar{a} \quad (59)$$

uniformly in i, u, θ . For every fixed $p < \infty$,

$$\sup_{k,\theta} \mathbb{E} \|G_{k,\theta}\|_2^p < \infty. \quad (60)$$

The trajectory log-likelihood Hessian $\mathcal{H}_{k,\theta}$ also satisfies

$$\sup_{k,\theta} \mathbb{E} \|\mathcal{H}_{k,\theta}\|_{\text{op}} < \infty. \quad (61)$$

Moreover, for every bounded measurable trajectory functional F , differentiation under the conditional trajectory law is valid up to order two:

$$\nabla_{\theta} \mathbb{E}_{\theta} F = \mathbb{E}_{\theta}(F G_{k,\theta}), \quad \nabla_{\theta}^2 \mathbb{E}_{\theta} F = \mathbb{E}_{\theta}\{F(G_{k,\theta} G_{k,\theta}^{\top} + \mathcal{H}_{k,\theta})\}. \quad (62)$$

Proof. Compactness of Θ and boundedness of $q_{i,u}$ imply (59). Every Dirichlet coordinate has a Beta marginal with parameters in a compact subset of $(0, \infty)^2$. Hence, for every finite p ,

$$\sup_{a,b} \int_0^1 |\log x|^p x^{a-1} (1-x)^{b-1} dx < \infty,$$

where the supremum is over the induced compact parameter set. The digamma and polygamma terms are bounded on the same positive compact interval. Equation (8), bounded features, and finite (h, N) therefore give (60); differentiating the score once more gives (61). The market kernel and $M_u(\cdot | s_u)$ contribute no likelihood derivative by Assumption 3.2.

To justify differentiation under the trajectory integral, enlarge Θ slightly while keeping it compactly contained in the policy domain. On this neighborhood the Dirichlet concentrations remain in $[a_-, a_+]$ for some $a_- > 0$. For one latent action $\tilde{\omega}$, the Dirichlet density and its first two θ -derivatives are bounded, up to constants independent of θ , by

$$\prod_{i=0}^N \tilde{\omega}_i^{a_- - 1} \left(1 + \sum_{i=0}^N |\log \tilde{\omega}_i|^2 \right).$$

This envelope is integrable on the simplex for $a_- > 0$. Taking the finite product over the h latent actions gives an integrable trajectory envelope; the parameter-independent market kernels integrate to one conditionally at each step. Hence bounded F times either derivative is dominated by a common integrable function. Dominated convergence yields (62). $\square$

A.3 Smoothness of the CPT objective

Lemma A.2 (Uniform gradient and smoothness bounds). *Under Assumptions 3.1–3.3, there exist $B_g, L < \infty$ such that*

$$\sup_{k,\theta \in \Theta} \|g_k(\theta)\|_2 \leq B_g, \quad \|g_k(\theta) - g_k(\theta')\|_2 \leq L \|\theta - \theta'\|_2. \quad (63)$$

With

$$C_{G,2} := \sup_{k,\theta} \mathbb{E} \|G_{k,\theta}\|_2^2, \quad C_H := \sup_{k,\theta} \mathbb{E} \|\mathcal{H}_{k,\theta}\|_{\text{op}},$$

one admissible choice is

$$B_g = 2B_v C_1 \sqrt{C_{G,2}}, \quad L = 2B_v \{C_2 C_{G,2} + C_1 (C_{G,2} + C_H)\}. \quad (64)$$

Moreover $|J_k(\theta, r_k)| \leq B_v$.

Proof. For one side σ and threshold y , the likelihood-score identity and Lemma A.1 give

$$\nabla S_\theta^\sigma(y) = \mathbb{E}\{I_\sigma(y)G_\theta\}, \quad \nabla^2 S_\theta^\sigma(y) = \mathbb{E}\{I_\sigma(y)(G_\theta G_\theta^\top + \mathcal{H}_\theta)\}.$$

The bounded indicator and the moment bounds provide an integrable dominating function, so the derivative–expectation interchange is justified by dominated convergence. Consequently

$$\|\nabla S_\theta^\sigma(y)\|_2 \leq \sqrt{C_{G,2}}, \quad \|\nabla^2 S_\theta^\sigma(y)\|_{\text{op}} \leq C_{G,2} + C_H.$$

Differentiating the CPT integral twice and using Assumption 3.3 gives

$$\nabla^2 \int_0^{B_v} w_\sigma^\varepsilon(S_\theta^\sigma(y)) dy = \int_0^{B_v} \{(w_\sigma^\varepsilon)''(S) \nabla S \nabla S^\top + (w_\sigma^\varepsilon)'(S) \nabla^2 S\} dy.$$

The integrand is uniformly dominated by the constants above, giving (64). The first-derivative bound is analogous. Finally $0 \leq w_\sigma^\varepsilon \leq 1$ and the two integrals have common length B_v , so $|J_k| \leq B_v$. $\square$

A.4 Projected residual and constrained stationarity

Lemma A.3 (Projected residual characterizes first-order stationarity). *For closed convex Θ , $\gamma, \vartheta > 0$, and $g \in \mathbb{R}^d$,*

$$\Pi_\Theta\{\theta + \gamma d(g)\} = \theta \iff \langle g \rangle u - \theta \leq 0 \quad \forall u \in \Theta. \quad (65)$$

Hence the two definitions in (18) coincide.

Proof. The projection variational inequality gives $x = \Pi_\Theta(y)$ iff $\langle y - x \rangle u - x \leq 0$ for every $u \in \Theta$. Substitute $y = \theta + \gamma d(g)$ and use $d(g) = g / \max\{\|g\|_2, \vartheta\}$, a positive scalar multiple of g . $\square$

B Proofs for the Reference Mechanism and CPT Gradient

B.1 Same terminal wealth and path-dependent reference

Proof. It is enough to compute the two terminal references and then use strict monotonicity of the deterministic CPT value.

For path I, the first step is a gain, so its interim reference is

$$r_I^{(1)} = (1 - \eta_+)r_0 + \eta_+x_{\text{hi}}.$$

The first displayed condition in the proposition makes the second step a loss relative to this interim reference. Hence

$$r_I = (1 - \eta_-)\{(1 - \eta_+)r_0 + \eta_+x_{\text{hi}}\} + \eta_-x_f.$$

For path II, the first step is a loss and

$$r_{II}^{(1)} = (1 - \eta_-)r_0 + \eta_-x_{\text{lo}}.$$

The second displayed condition makes the move to x_f a gain, so

$$r_{II} = (1 - \eta_+)\{(1 - \eta_-)r_0 + \eta_-x_{\text{lo}}\} + \eta_+x_f.$$

Subtracting cancels the coefficient of r_0 and gives (34).

It remains to translate a nonzero reference gap into a CPT-value gap. A deterministic relative outcome y_0 has survival function equal to one below its corresponding value magnitude and zero above it. Since $w_\pm^\varepsilon(0) = 0$ and $w_\pm^\varepsilon(1) = 1$, its CPT value is $v_+(y) - v_-(y)$. The shifted-power gain term is strictly increasing on $y > 0$, and $-v_-(y)$ is strictly increasing on $y < 0$; the two pieces meet continuously at zero. Thus the signed deterministic CPT value is strictly increasing in y . If $r_I \neq r_{II}$, then $x_f - r_I \neq x_f - r_{II}$, so the two CPT values differ. $\square$

B.2 Reference contraction and CPT sensitivity

Proof. To prove (37), it is enough to establish three bounds: contraction of the reference recursion, Lipschitz continuity of the value transformation, and an L^1 survival-function coupling inequality.

Fix a wealth value x . The map $r \mapsto \Phi(r, x)$ is continuous, with slope $1 - \eta_+$ when $r \leq x$ and slope $1 - \eta_-$ when $r > x$. Splitting an interval at x when necessary therefore gives

$$|\Phi(r, x) - \Phi(r', x)| \leq (1 - \eta_{\min})|r - r'|.$$

Under the reference-isolated comparison specified in Theorem 5.2, the two trajectories have the same wealth path. Iterating the preceding inequality for h steps proves (36). Since their terminal wealth is common,

$$|Y(r) - Y(r')| \leq (1 - \eta_{\min})^h |r - r'|. \quad (66)$$

For $x \geq 0$,

$$\frac{d}{dx} u_{\alpha, \varepsilon_v}(x) = \alpha x(x^2 + \varepsilon_v^2)^{\alpha/2-1} \leq \alpha \varepsilon_v^{\alpha-1},$$

where the bound uses $x \leq (x^2 + \varepsilon_v^2)^{1/2}$ and $\alpha \leq 1$. Hence

$$|v_+(y) - v_+(y')| + |v_-(y) - v_-(y')| \leq L_v |y - y'|.$$

Combining this with (66) controls the sum of the two terminal value-magnitude differences.

Finally, if $V, V' \in [0, B_v]$ are coupled nonnegative random variables, Tonelli's theorem gives

$$\begin{aligned} \int_0^{B_v} |P(V > y) - P(V' > y)| dy &\leq \mathbb{E} \int_0^{B_v} |\mathbf{1}\{V > y\} - \mathbf{1}\{V' > y\}| dy \\ &= \mathbb{E}|V - V'|. \end{aligned}$$

Since $|w_\sigma^\varepsilon(p) - w_\sigma^\varepsilon(q)| \leq C_1 |p - q|$, applying this inequality to the gain and loss integrals and using the preceding value bound proves (37). The decomposition (38) follows by adding and subtracting $J(P_{k+1}, z_{k+1}, r_k; \theta)$ and applying the triangle inequality uniformly in θ . $\square$

B.3 Score representation of the CPT gradient

Proof. We prove the gain part; the loss part follows by the same calculation and is subtracted at the end. To differentiate

$$J^+(\theta) = \int_0^{B_v} w_+^\varepsilon(S_\theta^+(y)) dy,$$

it is enough to identify $\nabla S_\theta^+(y)$ and justify moving the derivative through the y -integral.

For fixed y , the likelihood-ratio identity in Theorem A.1 applied to the bounded indicator gives

$$\nabla S_\theta^+(y) = \mathbb{E}\{\mathbf{1}[v_+(Y) > y] G_\theta\}. \quad (67)$$

By Cauchy-Schwarz and Assumption 3.1, the norm of the right-hand side is at most $\sqrt{C_{G,2}}$ uniformly in y and θ . Since $(w_+^\varepsilon)'$ is bounded by C_1 and the integration interval is finite, this constant is an integrable dominating function. The dominated convergence theorem therefore gives

$$\begin{aligned} \nabla J^+(\theta) &= \int_0^{B_v} (w_+^\varepsilon)'(S_\theta^+(y)) \mathbb{E}\{\mathbf{1}[v_+(Y) > y] G_\theta\} dy \\ &= \mathbb{E} \left[G_\theta \int_0^{B_v} (w_+^\varepsilon)'(S_\theta^+(y)) \mathbf{1}[v_+(Y) > y] dy \right], \end{aligned}$$

where Fubini's theorem is applicable because the integrand is dominated by $C_1 B_v \|G_\theta\|$ and $\mathbb{E}\|G_\theta\| < \infty$.

The score has mean zero. Therefore subtracting the deterministic quantity $S_\theta^+(y)$ inside the integral does not change the expectation. This yields the gain component of (39). Repeating the argument for the loss side and subtracting proves (40). The frozen-simulator identity follows under the same conditional argument with its trajectory law in place of the true law. $\square$

C Proofs for the Debiased Estimator

Throughout this appendix, condition on one frozen target and write $g = \nabla J(\theta)$. The complete trajectory summaries $\mathbf{X}_1, \mathbf{X}_2, \dots$ are independent and identically distributed under this conditional law. For a realization $x = (V^+, V^-, G_x)$, write $I_{x,\sigma}(y) = \mathbf{1}\{V^\sigma > y\}$.

C.1 Second-order expansion and full/half coupling

Lemma C.1 (Coupled full/half cancellation). *Under Assumptions 3.1–3.3, fix one frozen target and parameter and write $g := \nabla J(\theta)$. There exist a measurable $\mathfrak{h}$ and constants $C_{\text{rem}}, C_U < \infty$ such that*

$$U_n^{\text{raw}} - g = \frac{1}{n} \sum_{i=1}^n \mathfrak{h}(\mathbf{X}_i) + \mathcal{B}_n, \quad \mathbb{E}\mathfrak{h}(\mathbf{X}_i) = 0, \quad \mathbb{E}\|\mathcal{B}_n\|_2^2 \leq \frac{C_{\text{rem}}}{n^2}, \quad (68)$$

and consequently

$$\mathbb{E}\|D_{n,\ell}^{\text{raw}}\|_2^2 \leq \frac{9C_{\text{rem}}}{n^2} 2^{-2\ell}, \quad \ell \geq 1, \quad (69)$$

while

$$\text{tr Cov}(U_n^{\text{cen}}) \leq \frac{C_U}{n}. \quad (70)$$

Proof of Theorem C.1. To prove (68), it is enough to isolate the first-order Hoeffding projection and show that both the degenerate two-sample term and the Taylor remainder are $O_{L^2}(n^{-1})$. For a side σ and threshold y , write

$$S_\sigma(y) = P(V^\sigma > y), \quad e_{-i,\sigma}(y) = \widehat{S}_{-i,n}^\sigma(y) - S_\sigma(y).$$

Taylor's theorem applied to $(w_\sigma^\varepsilon)'$ gives

$$(w_\sigma^\varepsilon)'(S_\sigma + e) = (w_\sigma^\varepsilon)'(S_\sigma) + (w_\sigma^\varepsilon)''(S_\sigma)e + \text{Rem}_\sigma(e), \quad |\text{Rem}_\sigma(e)| \leq \frac{C_3}{2} e^2. \quad (71)$$

Substituting (71) into (28) splits the statistic into a first-order trajectory term, an ordered two-sample term, and a Taylor remainder.

Define

$$\begin{aligned} h_0(x) &= G_x \sum_{\sigma} \varsigma_{\sigma} \int_0^{B_v} (w_\sigma^\varepsilon)'(S_\sigma(y)) I_{x,\sigma}(y) dy, \\ \mathcal{K}(x, x') &= G_x \sum_{\sigma} \varsigma_{\sigma} \int_0^{B_v} (w_\sigma^\varepsilon)''(S_\sigma(y)) I_{x,\sigma}(y) \{I_{x',\sigma}(y) - S_\sigma(y)\} dy. \end{aligned}$$

By (40) and $\mathbb{E}G = 0$, $\mathbb{E}h_0(\mathbf{X}) = g$. Moreover $\mathbb{E}\{\mathcal{K}(\mathbf{X}, \mathbf{X}') \mid \mathbf{X}\} = 0$ for independent $\mathbf{X}, \mathbf{X}'$, because $\mathbb{E}\{I_{\mathbf{X}',\sigma}(y) - S_\sigma(y)\} = 0$. Let

$$h_1(x') = \mathbb{E}_{\mathbf{X}}\{\mathcal{K}(\mathbf{X}, x')\}.$$

Then $\mathbb{E}h_1(\mathbf{X}') = 0$. Put $\mathcal{K}^\circ(x, x') = \mathcal{K}(x, x') - h_1(x')$. Both first-order projections of $\mathcal{K}^\circ$ vanish:

$$\mathbb{E}\{\mathcal{K}^\circ(\mathbf{X}, \mathbf{X}') \mid \mathbf{X}\} = 0, \quad \mathbb{E}\{\mathcal{K}^\circ(\mathbf{X}, \mathbf{X}') \mid \mathbf{X}'\} = 0.$$

The ordered two-sample term therefore has the exact decomposition

$$\frac{1}{n(n-1)} \sum_{i \neq j} \mathcal{K}(\mathbf{X}_i, \mathbf{X}_j) = \frac{1}{n} \sum_{j=1}^n h_1(\mathbf{X}_j) + \mathcal{D}_n, \quad (72)$$

where

$$\mathcal{D}_n = \frac{1}{n(n-1)} \sum_{i \neq j} \mathcal{K}^\circ(\mathbf{X}_i, \mathbf{X}_j).$$

The kernel is square-integrable because the indicators are bounded, the integration interval is finite, $(w_\sigma^\varepsilon)''$ is bounded, and $\mathbb{E}\|G\|_2^2 < \infty$. Degeneracy implies that covariances between two ordered kernel terms vanish unless their ordered index pairs coincide or are reversals of one another. Counting these $O(n^2)$ contributing pairs against the normalization $n^2(n-1)^2$ gives

$$\mathbb{E}\|\mathcal{D}_n\|_2^2 \leq \frac{C_D}{n^2} \quad (73)$$

for a finite constant C_D .

We next bound the Taylor remainder. For each i , $e_{-i,\sigma}$ is an empirical mean based on $n-1$ Bernoulli variables independent of $\mathbf{X}_i$. Uniformly in y ,

$$\mathbb{E}|e_{-i,\sigma}(y)|^4 \leq \frac{C_e}{(n-1)^2} \quad (74)$$

for a universal finite C_e . By (71), Cauchy–Schwarz in the y -integral, independence of G_i from the leave-one-out empirical survival, and $\mathbb{E}\|G_i\|_2^2 < \infty$, the i th Taylor contribution has second moment at most C_T/n^2 . Jensen’s inequality for the average over i then gives a second-moment bound of the same order for the averaged Taylor remainder T_n :

$$\mathbb{E}\|T_n\|_2^2 \leq \frac{C'_T}{n^2}. \quad (75)$$

Combining the first-order term, (72), and the Taylor remainder yields

$$U_n^{\text{raw}} - g = \frac{1}{n} \sum_{i=1}^n \mathfrak{h}(\mathbf{X}_i) + \mathcal{B}_n, \quad \mathfrak{h} = h_0 - g + h_1, \quad \mathcal{B}_n = \mathcal{D}_n + T_n.$$

The vector function $\mathfrak{h}$ has mean zero, and (73)–(75) imply $\mathbb{E}\|\mathcal{B}_n\|_2^2 \leq C_{\text{rem}}/n^2$ after increasing the finite constant C_{rem} . This proves (68).

Now take $n_\ell = n2^\ell$ and evaluate the full-sample statistic and its two half-sample statistics on the same n_ℓ trajectories. Applying (68) to the full sample and to each half gives

$$U_{n_\ell}^{\text{raw}} = g + \frac{1}{n_\ell} \sum_{i=1}^{n_\ell} \mathfrak{h}(\mathbf{X}_i) + \mathcal{B}_{n_\ell},$$

and

$$\frac{1}{2}\{U_{n_\ell/2}^{\text{raw},(1)} + U_{n_\ell/2}^{\text{raw},(2)}\} = g + \frac{1}{n_\ell} \sum_{i=1}^{n_\ell} \mathfrak{h}(\mathbf{X}_i) + \frac{1}{2}\{\mathcal{B}_{n_\ell/2}^{(1)} + \mathcal{B}_{n_\ell/2}^{(2)}\}.$$

The influence terms cancel exactly. Hence

$$D_{n,\ell}^{\text{raw}} = \mathcal{B}_{n_\ell} - \frac{1}{2} \{ \mathcal{B}_{n_\ell/2}^{(1)} + \mathcal{B}_{n_\ell/2}^{(2)} \}.$$

Minkowski's inequality in L^2 gives

$$(\mathbb{E} \|D_{n,\ell}^{\text{raw}}\|_2^2)^{1/2} \leq \frac{\sqrt{C_{\text{rem}}}}{n_\ell} + \frac{1}{2} \frac{2\sqrt{C_{\text{rem}}}}{n_\ell} + \frac{1}{2} \frac{2\sqrt{C_{\text{rem}}}}{n_\ell} = \frac{3\sqrt{C_{\text{rem}}}}{n_\ell}.$$

Squaring and substituting $n_\ell = n2^\ell$ proves (69).

It remains to prove (70). View U_n^{cen} as a symmetric function of $\mathbf{X}_1, \dots, \mathbf{X}_n$, and let $U_n^{\text{cen},(i)}$ be the statistic after replacing $\mathbf{X}_i$ by an independent copy $\mathbf{X}'_i$. If G'_i denotes the score component of $\mathbf{X}'_i$, write $L_f = C_1 + C_2$. The coefficient in one centered LOO summand is bounded in absolute value by $2B_v C_1$, so the contribution from the replaced summand has norm at most

$$\frac{2B_v C_1}{n} (\|G_i\|_2 + \|G'_i\|_2).$$

For $j \neq i$, replacing $\mathbf{X}_i$ changes each leave-one-out survival entering the j th summand by at most $(n-1)^{-1}$ pointwise. Since

$$\left| \frac{\partial}{\partial p} \{ (w_\sigma^\varepsilon)'(p)(I-p) \} \right| \leq L_f,$$

the scalar coefficient multiplying G_j changes by at most $2B_v L_f / (n-1)$. Hence, for a deterministic constant C_* depending only on B_v, C_1, C_2 ,

$$\|U_n^{\text{cen}} - U_n^{\text{cen},(i)}\|_2 \leq \frac{C_*}{n} (\|G_i\|_2 + \|G'_i\|_2) + \frac{C_*}{n(n-1)} \sum_{j \neq i} \|G_j\|_2.$$

Using $(\sum_{j \neq i} x_j)^2 \leq (n-1) \sum_{j \neq i} x_j^2$ and the uniform second-moment bound for the score, there is a finite constant C_E such that

$$\mathbb{E} \|U_n^{\text{cen}} - U_n^{\text{cen},(i)}\|_2^2 \leq \frac{C_E}{n^2}$$

uniformly in i . The vector Efron–Stein inequality yields

$$\text{tr Cov}(U_n^{\text{cen}}) = \mathbb{E} \|U_n^{\text{cen}} - \mathbb{E} U_n^{\text{cen}}\|_2^2 \leq \frac{1}{2} \sum_{i=1}^n \mathbb{E} \|U_n^{\text{cen}} - U_n^{\text{cen},(i)}\|_2^2 \leq \frac{C_U}{n}$$

with $C_U = C_E/2$. This completes the proof. $\square$

C.2 Centering and finite-sample bias

Lemma C.2 (Centering and finite-sample bias). *For $n \geq 2$,*

$$\mathbb{E} U_n^{\text{cen}} = \mathbb{E} U_n^{\text{raw}} =: \bar{U}_n, \tag{76}$$

and there exists $C_{\text{bias}} < \infty$ such that

$$\|\bar{U}_n - g\|_2 \leq \frac{C_{\text{bias}}}{n}, \quad g := \nabla J(\theta). \tag{77}$$

Proof of Theorem C.2. To compare the two LOO statistics, subtract (27) from (28). The difference is

$$U_n^{\text{raw}} - U_n^{\text{cen}} = \frac{1}{n} \sum_{i=1}^n G_i H_{-i}^{\text{ctr}}, \quad (78)$$

where

$$H_{-i}^{\text{ctr}} = \sum_{\sigma \in \{+, -\}} \varsigma_\sigma \int_0^{B_v} (w_\sigma^\varepsilon)'(\hat{S}_{-i,n}^\sigma(y)) \hat{S}_{-i,n}^\sigma(y) dy.$$

The coefficient H_{-i}^{ctr} depends only on trajectories with indices different from i . It is therefore independent of G_i . The score identity gives $\mathbb{E}G_i = 0$, while Assumptions 3.1–3.3 make $G_i H_{-i}^{\text{ctr}}$ integrable. Hence every summand in (78) has mean zero and $\mathbb{E}U_n^{\text{cen}} = \mathbb{E}U_n^{\text{raw}} = \bar{U}_n$.

It remains to control $\bar{U}_n - g$. This follows from the expansion in Theorem C.1. Taking expectations in (68) and using $\mathbb{E}\mathbf{h}(\mathbf{X}) = 0$ yields

$$\|\bar{U}_n - g\|_2 = \|\mathbb{E}\mathcal{B}_n\|_2 \leq (\mathbb{E}\|\mathcal{B}_n\|_2^2)^{1/2} \leq \frac{\sqrt{C_{\text{rem}}}}{n}.$$

Thus (77) holds with $C_{\text{bias}} = \sqrt{C_{\text{rem}}}$. $\square$

C.3 Unbiasedness, variance, cost, and the CLT

Proof of Theorem 5.4. We first prove exact unbiasedness. By Theorem C.2, the centered base has mean $\bar{U}_n$. For the raw correction,

$$\mathbb{E}D_{n,\ell}^{\text{raw}} = \bar{U}_{n2^\ell} - \bar{U}_{n2^{\ell-1}}. \quad (79)$$

The level bound (69) implies

$$\sum_{\ell \geq 1} \mathbb{E}\|D_{n,\ell}^{\text{raw}}\|_2 \leq \frac{3\sqrt{C_{\text{rem}}}}{n} \sum_{\ell \geq 1} 2^{-\ell} < \infty,$$

so the expectation can be interchanged with the infinite randomized series. Since (77) gives $\bar{U}_{n2^{\ell_{\text{max}}}} \rightarrow g$,

$$\begin{aligned} \mathbb{E}Z_n^{\text{db}} &= \bar{U}_n + \sum_{\ell \geq 1} \mathbb{E}D_{n,\ell}^{\text{raw}} \\ &= \bar{U}_n + \lim_{\ell_{\text{max}} \rightarrow \infty} (\bar{U}_{n2^{\ell_{\text{max}}}} - \bar{U}_n) = g. \end{aligned}$$

This proves (42).

For the second moment of the correction, independence of the activation and level variables gives

$$\begin{aligned} \mathbb{E} \left\| \frac{\mathbf{A}}{p_{\text{act}} \nu_{\ell^*}} D_{n,\ell^*}^{\text{raw}} \right\|_2^2 &= \frac{1}{p_{\text{act}}} \sum_{\ell \geq 1} \frac{\mathbb{E}\|D_{n,\ell}^{\text{raw}}\|_2^2}{\nu_\ell} \\ &\leq \frac{9C_{\text{rem}}}{p_{\text{act}} n^2} \sum_{\ell \geq 1} \frac{4^{-\ell}}{\nu_\ell} = \frac{9C_{\text{rem}} \Xi_b}{p_{\text{act}} n^2}. \end{aligned}$$

The series Ξ_b is finite because $b < 2$. The base sample and correction sample are independent, so their covariance is zero. Combining the correction bound with (70) gives (43).

One base output costs exactly n complete trajectories. Conditional on activation and level ℓ , the correction uses one fresh batch of $n2^\ell$ trajectories; the two halves reuse this same batch. Hence

$$\mathbb{E} \text{cost}(Z_n^{\text{db}}) = n + p_{\text{act}} n \sum_{\ell \geq 1} \nu_\ell 2^\ell = n(1 + p_{\text{act}} \Upsilon_b),$$

where $\Upsilon_b < \infty$ because $b > 1$. Direct summation of the geometric series gives the closed forms in (41).

Finally, if $Z_{n,1}^{\text{db}}, \dots, Z_{n,M}^{\text{db}}$ are independent copies, exact unbiasedness and independence imply

$$\mathbb{E} \|\hat{g}_{M,n}^{\text{db}} - g\|_2^2 = \frac{1}{M} \text{tr Cov}(Z_n^{\text{db}}) \leq \frac{C_{\text{var}}(n)}{M}.$$

This is (45). □

Proof of Theorem 5.5. For fixed target and fixed (n, p_{act}, b) , Theorem 5.4 gives iid vectors $Z_{n,r}^{\text{db}}$ with mean g and finite covariance Σ_n . For any fixed $t \in \mathbb{R}^d$, the scalar variables $t^\top (Z_{n,r}^{\text{db}} - g)$ are iid with finite variance. The classical scalar central limit theorem therefore gives

$$\sqrt{M} t^\top (\hat{g}_{M,n}^{\text{db}} - g) \rightsquigarrow \mathcal{N}(0, t^\top \Sigma_n t).$$

The Cramér–Wold device proves (46).

Because $\mathbb{E} \|Z_n^{\text{db}}\|_2^2 < \infty$, every entry of $Z_n^{\text{db}} (Z_n^{\text{db}})^\top$ is integrable. The law of large numbers therefore implies $\hat{\Sigma}_{M,n} \rightarrow \Sigma_n$ in probability entrywise, hence in operator norm in fixed dimension. If Σ_n is nonsingular, operator-norm consistency implies $\mathbb{P}\{\lambda_{\min}(\hat{\Sigma}_{M,n}) > \lambda_{\min}(\Sigma_n)/2\} \rightarrow 1$. On this event the inverse square-root map is well defined and continuous in a neighborhood of Σ_n . Slutsky’s theorem applied to (46) then proves (47). □

D Proofs for the Online Residual and Tracking Results

D.1 Normalized projected ascent

Lemma D.1 (One-step normalized projected ascent). *Let J be L -smooth on the nonempty closed convex set Θ . Fix $x \in \Theta$, let $g = \nabla J(x)$ with $\|g\|_2 \leq B_g$, and let $\hat{g}$ be any finite vector. Set*

$$x^+ = \Pi_\Theta\{x + \gamma d(\hat{g})\}, \quad Q(x) = \gamma^{-1}[\Pi_\Theta\{x + \gamma d(g)\} - x].$$

If $\gamma L \leq \vartheta$ and $C_g := \max\{B_g, \vartheta\}$, then

$$J(x^+) - J(x) \geq \frac{\vartheta\gamma}{8} \|Q(x)\|_2^2 - \frac{5C_g\gamma}{4\vartheta^2} \|\hat{g} - g\|_2^2. \quad (80)$$

Proof of Theorem D.1. The target is a lower bound in terms of the population projected residual $Q(x)$. We first obtain a bound in terms of the actual displacement produced by $\hat{g}$, then compare that displacement with the population displacement.

Let

$$\Delta x = x^+ - x, \quad c_g = \max\{\|g\|_2, \vartheta\}, \quad \epsilon = d(\hat{g}) - d(g).$$

The projection variational inequality with the feasible comparison point x gives

$$\langle d(\hat{g}), \Delta x \rangle \geq \frac{\|\Delta x\|_2^2}{\gamma}. \quad (81)$$

Since $g = c_g \mathbf{d}(g)$, the L -smooth lower quadratic bound and (81) imply

$$\begin{aligned}
J(x + \Delta x) - J(x) &\geq \langle g, \Delta x \rangle - \frac{L}{2} \|\Delta x\|_2^2 \\
&= c_g \langle \mathbf{d}(\widehat{g}) - \mathbf{e}, \Delta x \rangle - \frac{L}{2} \|\Delta x\|_2^2 \\
&\geq \left(\frac{c_g}{\gamma} - \frac{L}{2} \right) \|\Delta x\|_2^2 - c_g \|\mathbf{e}\|_2 \|\Delta x\|_2 \\
&\geq \left(\frac{3c_g}{4\gamma} - \frac{L}{2} \right) \|\Delta x\|_2^2 - c_g \gamma \|\mathbf{e}\|_2^2 \\
&\geq \frac{c_g}{4\gamma} \|\Delta x\|_2^2 - c_g \gamma \|\mathbf{e}\|_2^2.
\end{aligned} \tag{82}$$

The penultimate line uses $uv \leq u^2/(4\gamma) + \gamma v^2$, and the last line uses $\gamma L \leq \vartheta \leq c_g$.

Now let

$$t = \Pi_{\Theta}\{x + \gamma \mathbf{d}(g)\} - x = \gamma Q(x).$$

Nonexpansiveness of Euclidean projection gives $\|\Delta x - t\|_2 \leq \gamma \|\mathbf{e}\|_2$. From $\|t\|_2^2 \leq 2\|\Delta x\|_2^2 + 2\|\Delta x - t\|_2^2$,

$$\|\Delta x\|_2^2 \geq \frac{1}{2} \|t\|_2^2 - \gamma^2 \|\mathbf{e}\|_2^2.$$

Substituting into (82) yields

$$J(x^+) - J(x) \geq \frac{c_g \gamma}{8} \|Q(x)\|_2^2 - \frac{5c_g \gamma}{4} \|\mathbf{e}\|_2^2.$$

Finally, $\mathbf{d}(g) = \Pi_{\{u: \|u\|_2 \leq 1\}}(g/\vartheta)$, so nonexpansiveness of the unit-ball projection implies

$$\|\mathbf{e}\|_2 \leq \vartheta^{-1} \|\widehat{g} - g\|_2.$$

Using $c_g \geq \vartheta$ in the positive term and $c_g \leq C_g = \max(B_g, \vartheta)$ in the negative term proves (80). $\square$

D.2 Average expected squared projected residual

Proof of Theorem 5.8. We first connect the final estimator to the true gradient. Conditional on $\mathcal{F}_k$, Theorem 5.4 gives

$$\mathbb{E}(\widehat{g}_k^{\text{db}} - \widetilde{g}_k(\theta_k) \mid \mathcal{F}_k) = 0, \quad \mathbb{E}(\|\widehat{g}_k^{\text{db}} - \widetilde{g}_k(\theta_k)\|_2^2 \mid \mathcal{F}_k) \leq \frac{C_{\text{var}}(n)}{M_k}.$$

The cross term with the deterministic conditional vector $\widetilde{g}_k(\theta_k) - g_k(\theta_k)$ therefore vanishes. Hence

$$\begin{aligned}
\mathbb{E}\|\widehat{g}_k^{\text{db}} - g_k(\theta_k)\|_2^2 &= \mathbb{E}\|\widehat{g}_k^{\text{db}} - \widetilde{g}_k(\theta_k)\|_2^2 + \mathbb{E}\|\widetilde{g}_k(\theta_k) - g_k(\theta_k)\|_2^2 \\
&\leq \frac{C_{\text{var}}(n)}{M_k} + \mathbb{E}\beta_k^2,
\end{aligned}$$

which is (51).

We next apply Theorem D.1 to the true episode objective J_k at $x = \theta_k$ and $\widehat{g} = \widehat{g}_k^{\text{db}}$. Let

$$\mathbf{j}_k = J_k(\theta_{k+1}) - J_k(\theta_k).$$

After taking expectations,

$$\frac{\vartheta\gamma}{8}\mathbb{E}\zeta_k \leq \mathbb{E}\mathbf{j}_k + \frac{5C_g\gamma}{4\vartheta^2}\mathbb{E}\|\hat{g}_k^{\text{db}} - g_k(\theta_k)\|_2^2. \quad (83)$$

The objective increments telescope across moving objectives as the pathwise identity

$$\begin{aligned} \sum_{k=1}^K \mathbf{j}_k &= J_K(\theta_{K+1}) - J_1(\theta_1) \\ &\quad + \sum_{k=1}^{K-1} \{J_k(\theta_{k+1}) - J_{k+1}(\theta_{k+1})\}. \end{aligned} \quad (84)$$

Since $|J_k| \leq B_v$ and the second line is bounded above by $\mathcal{V}_K$,

$$\sum_{k=1}^K \mathbf{j}_k \leq 2B_v + \mathcal{V}_K. \quad (85)$$

Summing (83), applying (51) and (85), and dividing by $\vartheta\gamma K/8$ gives exactly (52). $\square$

D.3 Dynamic local regret

Proof of Theorem 5.9. Write $A = A_{\rho,W}$. With the convention $Q_t(\theta_t) = 0$ for $t \leq 0$, the coefficients ρ^j/A , $j = 0, \dots, W-1$, are nonnegative and sum to one. Convexity of $x \mapsto \|x\|_2^2$ gives

$$\|\bar{Q}_k\|_2^2 \leq \frac{1}{A} \sum_{j=0}^{W-1} \rho^j \|Q_{k-j}(\theta_{k-j})\|_2^2.$$

Summing over $k \leq K$ and reindexing,

$$\mathcal{R}_{\rho,W}(K) \leq \sum_{t=1}^K \zeta_t \left\{ \frac{1}{A} \sum_{j=0}^{\min\{W-1, K-t\}} \rho^j \right\} \leq \sum_{t=1}^K \zeta_t.$$

This proves (53); taking expectations and dividing by K gives (54). $\square$

Independent-stream evaluation. For a fixed iterate θ , write

$$\mathcal{Q}_\theta(g) := \gamma^{-1}[\Pi_\Theta\{\theta + \gamma \mathbf{d}(g)\} - \theta].$$

Projection nonexpansiveness and the ϑ^{-1} -Lipschitz property of $\mathbf{d}$ imply

$$\|\mathcal{Q}_\theta(\hat{g}) - \mathcal{Q}_\theta(g)\|_2 \leq \vartheta^{-1}\|\hat{g} - g\|_2. \quad (86)$$

Hence, if $\hat{g}^{(1)}, \hat{g}^{(2)}$ are independent unbiased gradient estimates with $\mathbb{E}\|\hat{g}^{(j)} - g\|_2^2 \leq \varepsilon_g^2$, and $\hat{Q}^{(j)} = \mathcal{Q}_\theta(\hat{g}^{(j)})$, then

$$\left| \mathbb{E}\langle \hat{Q}^{(1)} | \hat{Q}^{(2)} \rangle - \|\mathcal{Q}_\theta(g)\|_2^2 \right| \leq \frac{2\varepsilon_g}{\vartheta} + \frac{\varepsilon_g^2}{\vartheta^2}. \quad (87)$$

Indeed, independence gives $\mathbb{E}\langle \hat{Q}^{(1)} | \hat{Q}^{(2)} \rangle = \|\mathbb{E}\hat{Q}^{(1)}\|_2^2$. Writing $e = \mathbb{E}\hat{Q}^{(1)} - \mathcal{Q}_\theta(g)$, Jensen's inequality and (86) give $\|e\|_2 \leq \varepsilon_g/\vartheta$. Finally $\|\mathcal{Q}_\theta(g)\|_2 \leq 1$, because $\|\mathbf{d}(g)\|_2 \leq 1$, which yields (87).

The same argument passes through the linear DLR smoothing. If

$$\widehat{\bar{Q}}_k^{(j)} = A_{\rho, W}^{-1} \sum_{r=0}^{W-1} \rho^r \widehat{Q}_{k-r}^{(j)}, \quad \bar{\varepsilon}_k = A_{\rho, W}^{-1} \sum_{r=0}^{W-1} \rho^r \varepsilon_{g, k-r},$$

with prehistory terms set to zero as in (20), then

$$\left| \mathbb{E} \langle \widehat{\bar{Q}}_k^{(1)} \rangle \widehat{\bar{Q}}_k^{(2)} - \|\bar{Q}_k\|_2^2 \right| \leq \frac{2\bar{\varepsilon}_k}{\vartheta} + \frac{\bar{\varepsilon}_k^2}{\vartheta^2}. \quad (88)$$

D.4 Stationary-set consequences

Lemma D.2 (Drift of stationary sets). *If (55) holds, define*

$$\delta_k^g := \sup_{\theta \in \Theta} \|g_{k+1}(\theta) - g_k(\theta)\|_2. \quad (89)$$

Then

$$d_H(\mathbf{S}_{k+1}, \mathbf{S}_k) \leq \frac{\kappa}{\vartheta} \delta_k^g. \quad (90)$$

Proof of Theorem 5.10. Condition (55) gives pointwise

$$\text{dist}^2(\theta_k, \mathbf{S}_k) \leq \kappa^2 \|Q_k(\theta_k)\|_2^2 = \kappa^2 \zeta_k.$$

Taking expectations and averaging over k proves (56). Substitution of (52) yields the stated finite-horizon control. $\square$

Proof of Theorem D.2. The normalized direction map is ϑ^{-1} -Lipschitz because it is the Euclidean projection of g/ϑ onto the closed unit ball. Using nonexpansiveness of Π_Θ in (17),

$$\begin{aligned} \sup_{\theta \in \Theta} \|Q_{k+1}(\theta) - Q_k(\theta)\|_2 &\leq \sup_{\theta \in \Theta} \|\mathbf{d}(g_{k+1}(\theta)) - \mathbf{d}(g_k(\theta))\|_2 \\ &\leq \frac{\delta_k^g}{\vartheta}. \end{aligned}$$

If $\theta \in \mathbf{S}_k$, then $Q_k(\theta) = 0$, so (55) applied to episode $k+1$ gives

$$\text{dist}(\theta, \mathbf{S}_{k+1}) \leq \kappa \|Q_{k+1}(\theta)\|_2 \leq \frac{\kappa \delta_k^g}{\vartheta}.$$

Taking the supremum over $\theta \in \mathbf{S}_k$ gives one directed Hausdorff bound. Interchanging k and $k+1$ gives the reverse bound because the same uniform gradient-difference bound is symmetric. Taking the maximum proves (90). $\square$

Proof of Theorem 5.11. In (52), the endpoint term is $O(K^{-1})$. The assumption $\mathbb{E}\mathcal{V}_K = o(K)$ makes the variation term vanish after division by K . Since $C_{\text{var}}(n)$ is fixed,

$$\frac{1}{K} \sum_{k=1}^K \left(\frac{C_{\text{var}}(n)}{M_k} + \mathbb{E}\beta_k^2 \right) \rightarrow 0$$

by the assumed Cesàro condition. Therefore $\mathcal{E}_K \rightarrow 0$. Equation (54) gives the DLR conclusion, and (56) gives the stationary-set conclusion when the local condition (55) holds. $\square$

Objective variation in the semi-synthetic environment. Let c_k^{ret} collect the episode- k coefficients of the semi-synthetic return law. The E3 construction makes the episode-start account changes satisfy

$$|\log X_{t_{k+1}} - \log X_{t_k}| + |r_{k+1} - r_k| + \|\omega_{t_{k+1}-1} - \omega_{t_k-1}\|_1 = O(\varpi_k), \quad \varpi_k = k^{-3/4},$$

while c_k^{ret} changes only at the regime boundary and over a finite interpolation, so $\sum_k \|c_{k+1}^{\text{ret}} - c_k^{\text{ret}}\| < \infty$. For a frozen objective that is uniformly Lipschitz in these episode-start inputs on the compact analysis domain, there are finite constants $L_{\text{acct}}, L_{\text{ret}}$ such that

$$\begin{aligned} \sup_{\theta \in \Theta} |J_{k+1}(\theta, r_{k+1}) - J_k(\theta, r_k)| &\leq L_{\text{acct}} (|\log X_{t_{k+1}} - \log X_{t_k}| + |r_{k+1} - r_k| \\ &\quad + \|\omega_{t_{k+1}-1} - \omega_{t_k-1}\|_1) + L_{\text{ret}} \|c_{k+1}^{\text{ret}} - c_k^{\text{ret}}\|. \end{aligned}$$

Under this regularity, the schedule gives $\mathcal{V}_K = O(\sum_{k < K} \varpi_k) + O(1) = O(K^{1/4}) = o(K)$. The experiment uses the same conditional factor-block and return law for training and execution, so the simulator discrepancy term is zero at the population level for this semi-synthetic construction.

E Proof of the Fixed-Dimension Minimax Rate

Proof of Theorem 5.7. We prove the lower and upper bounds under the same complete-trajectory oracle and the same expected trajectory budget. For the lower bound, it is enough to construct a two-point submodel contained in $\mathfrak{C}$ whose CPT gradients are separated by order $B^{-1/2}$ while the stopped transcript distributions remain at bounded total variation. For the upper bound, Theorem 5.4 supplies an unbiased finite-variance output with finite expected trajectory cost.

Step 1: a CPT-realizable one-stock submodel. Consider one risky asset and cash, horizon $h = 1$, zero transaction cost, and episode-start wealth and reference equal to one. Take the previous portfolio to be all cash and let the feasibility map be the identity. Set the risky feature to $e_1 \in \mathbb{R}^d$ and the cash feature to zero. At a query θ , the latent risky weight satisfies

$$U \sim \text{Beta}(e^{\theta^{[1]}}, 1),$$

where $\theta^{[1]}$ is the first coordinate of θ . After the inertia map (7), the executed risky weight is χU .

Let the risky return be $R = R_\star \xi$, where $R_\star > 0$ is fixed and sufficiently small and $\xi \sim \text{Bernoulli}(p)$ is independent of U . The terminal wealth and reference are

$$X_{\text{end}} = 1 + \chi R_\star U \xi, \quad r_{\text{end}} = 1 + \eta_+ \chi R_\star U \xi,$$

so the terminal relative outcome is

$$Y = v U \xi, \quad v = (1 - \eta_+) \chi R_\star > 0. \tag{91}$$

This is an admissible portfolio trajectory model under Definition 5.6. The loss-side value is identically zero.

Step 2: local separation of the CPT gradient. Let $v = v_+$ and define

$$\mathbf{m}(t) = v v'(vt), \quad 0 \leq t \leq 1.$$

The shifted-power value function gives $\mathbf{m}(0) = 0$, $\mathbf{m}(t) > 0$ for $t \in (0, 1]$, and bounded $\mathbf{m}$. Since $P(\mathbf{U} > t) = 1 - t^{e^{\theta^{[1]}}}$, a change of variables $y = v(vt)$ in the gain-side CPT integral gives

$$J_p(\theta) = \int_0^1 \mathbf{m}(t) w_+^\varepsilon \left(p \{1 - t^{e^{\theta^{[1]}}}\} \right) dt. \quad (92)$$

At the target $\theta_0 = 0$, differentiation under the integral is valid because, on a compact neighborhood of θ_0 , $t^{e^{\theta^{[1]}}} |\log t|$ has an integrable envelope on $(0, 1)$ and $(w_+^\varepsilon)'$ is bounded. Therefore

$$\nabla J_p(0) = \mathbf{g}(p) e_1, \quad \mathbf{g}(p) = p \int_0^1 \mathbf{m}(t) (w_+^\varepsilon)'(p(1-t)) (-t \log t) dt. \quad (93)$$

Put

$$A_v = \int_0^1 \mathbf{m}(t) (-t \log t) dt > 0, \quad d_W = (w_+^\varepsilon)'(0) > 0.$$

Since $|(w_+^\varepsilon)''| \leq C_2$, differentiation of (93) with respect to p gives

$$\begin{aligned} \mathbf{g}'(p) &= \int_0^1 \mathbf{m}(t) (-t \log t) \left[(w_+^\varepsilon)'(p(1-t)) + p(1-t) (w_+^\varepsilon)''(p(1-t)) \right] dt \\ &\geq (d_W - 2C_2 p) A_v. \end{aligned}$$

Choose $p_{\max} \in (0, 1/2]$ small enough that $2C_2 p_{\max} \leq d_W/2$; if $C_2 = 0$, any choice in $(0, 1/2]$ is admissible. Let

$$p_* = p_{\max}/2, \quad \mathcal{I} = [p_*/2, 3p_*/2], \quad c_* = d_W A_v/2.$$

Then

$$\mathbf{g}'(p) \geq c_* > 0 \quad \text{for every } p \in \mathcal{I}. \quad (94)$$

Step 3: information in an adaptive stopped transcript. For B large enough, take

$$p_\pm = p_* \pm \delta, \quad \delta = \delta_*/\sqrt{B},$$

where $\delta_* > 0$ is chosen so that $p_\pm \in \mathcal{I}$. For Bernoulli laws in the fixed interior interval $\mathcal{I}$, the elementary bound $\text{kl}(p\|q) \leq (p-q)^2/\{q(1-q)\}$ gives

$$\text{kl}(p_+\|p_-) \leq C_I \delta^2 \quad (95)$$

for a finite constant C_I depending only on $\mathcal{I}$.

Consider an arbitrary admissible estimator in (49). Its query point at each oracle call may depend on the preceding transcript. Conditional on any such query, the latent Dirichlet action law is identical under p_+ and p_- ; only the Bernoulli return kernel differs. All algorithmic randomization can be included in the transcript and is likewise common to the two models. The data-dependent call count T_{call} is a stopping time for the oracle transcript filtration. The chain rule for the stopped likelihood ratio gives

$$\text{KL}(\mathbb{P}_+^{\text{tr}}\|\mathbb{P}_-^{\text{tr}}) = \mathbb{E}_+ T_{\text{call}} \text{kl}(p_+\|p_-) \leq B C_I \delta^2 = C_I \delta_*^2. \quad (96)$$

The equality can be obtained by summing the conditional log-likelihood increments over $\{T_{\text{call}} \geq j\}$; these events are measurable before the j th fresh return is observed. The increments are uniformly integrable because $p_\pm$ stay in the compact interior interval $\mathcal{I}$ and $\mathbb{E}_+ T_{\text{call}} < \infty$.

Choose δ_* small enough that $C_I \delta_*^2 \leq 1/8$. Pinsker's inequality then gives

$$\text{TV}(\mathbb{P}_+^{\text{tr}}, \mathbb{P}_-^{\text{tr}}) \leq \frac{1}{4}. \quad (97)$$

Step 4: reduction from estimation to testing. By (94) and the mean-value theorem, the two gradient targets satisfy

$$D_{\mathbf{B}}^g := \|\mathbf{g}(p_+)e_1 - \mathbf{g}(p_-)e_1\|_2 \geq 2c_*\delta. \quad (98)$$

Given any estimator $\hat{g}$, define the test that chooses the closer of the two targets. On a testing error, the estimator must be at least $D_{\mathbf{B}}^g/2$ from the true target. The sum of the two testing error probabilities is at least one minus the total variation distance. Therefore

$$\begin{aligned} \max_{s \in \{+, -\}} \mathbb{E}_s \|\hat{g} - \mathbf{g}(p_s)e_1\|_2^2 &\geq \frac{(D_{\mathbf{B}}^g)^2}{8} \{1 - \text{TV}(\mathbb{P}_+^{\text{tr}}, \mathbb{P}_-^{\text{tr}})\} \\ &\geq \frac{3c_*^2\delta_*^2}{8\mathbf{B}}. \end{aligned}$$

The two models belong to $\mathfrak{C}$, so taking the infimum over all admissible procedures gives the lower bound in (50) with $c_{\text{lb}} = 3c_*^2\delta_*^2/8$.

Step 5: cost-matched upper bound. Fix any admissible estimator design parameters (n, p_{act}, b) from Theorem 5.4. The common model-class constants in Definition 5.6 make

$$\sup_{P \in \mathfrak{C}} \text{tr Cov}_P(Z_n^{\text{db}}) \leq C_{\text{var}}(n) < \infty, \quad C_{\text{cost}} = n(1 + p_{\text{act}}\Upsilon_b) < \infty$$

uniform over $P \in \mathfrak{C}$. For

$$M = \left\lfloor \frac{\mathbf{B}}{C_{\text{cost}}} \right\rfloor,$$

run M independent, untruncated debiased outputs Z_n^{db} at θ_0 and average them. Their total expected number of complete trajectories is $MC_{\text{cost}} \leq \mathbf{B}$, so this is an admissible procedure. For $\mathbf{B} \geq 2C_{\text{cost}}$, $M \geq \mathbf{B}/(2C_{\text{cost}})$. Hence

$$\sup_{P \in \mathfrak{C}} \mathbb{E}_P \|\hat{g}_{M,n}^{\text{db}} - g_P\|_2^2 \leq \frac{C_{\text{var}}(n)}{M} \leq \frac{2C_{\text{var}}(n)C_{\text{cost}}}{\mathbf{B}}.$$

Thus the upper bound in (50) holds with $C_{\text{ub}} = 2C_{\text{var}}(n)C_{\text{cost}}$ and a sufficiently large $\mathbf{B}_0$. The lower and upper bounds use the same target, trajectory oracle, loss, and expected raw-trajectory budget, completing the proof. $\square$